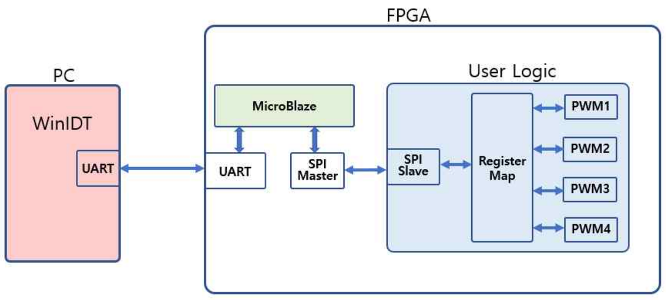
SoC 를 위한 *Peripheral* 설계

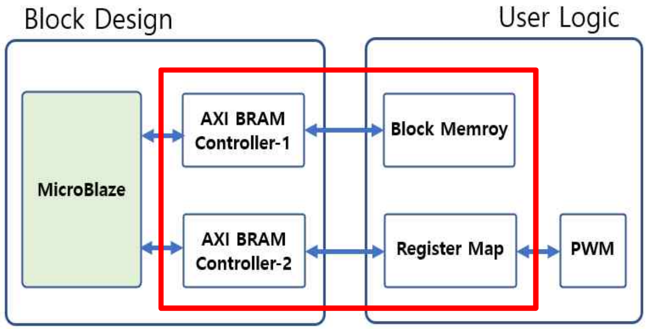
[*Create and Package New IP* (I , II)]

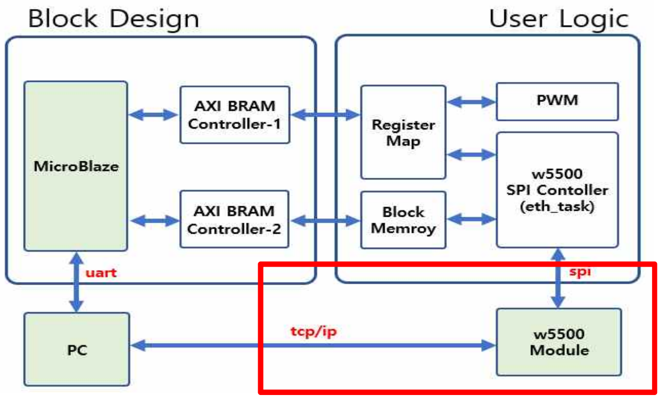
Reference :

Table of Contents

➢ SoC 를 위한 Peripheral 설계







➢ SoC Peripheral RTC Design Project

[*Create and Package New IP (I)*]

SPI_Task and Top_FndController

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢ U0 : spi_task(spi_task.v)

spi_task(spi_task.v)(1/4)

```
`timescale 1ns / 1ps

module spi_task(reset   , clock   , btn     , led
);
input                                   reset   ;
input                                   clock   ;
input           [3:0]                   btn     ;
output          [7:0]                   led     ;

wire            [3:0]                   btn2 ;
btn_in          btn_in_u1 (
                .reset   (reset   ),
                .clock   (clock   ),
                .btn_in  (btn[0]  ),
                .btn_out (btn2[0] )
);
btn_in          btn_in_u2 (
                .reset   (reset   ),
                .clock   (clock   ),
                .btn_in  (btn[1]  ),
                .btn_out (btn2[1] )
);
btn_in          btn_in_u3 (
                .reset   (reset   ),
                .clock   (clock   ),
                .btn_in  (btn[2]  ),
                .btn_out (btn2[2] )
);
btn_in          btn_in_u4 (
                .reset   (reset   ),
                .clock   (clock   ),
                .btn_in  (btn[3]  ),
                .btn_out (btn2[3] )
);

reg             [1:0]                   btn_id;
always @(posedge clock or negedge reset)
begin
                if(~reset)              btn_id <= 2'b0;
                else                    btn_id <= btn2[0] ? 2'b00 :
                                btn2[1] ? 2'b01 :
                                btn2[2] ? 2'b10 :
                                btn2[3] ? 2'b11 : btn_id;
end

reg             [7:0]                   wdata0;
always @(posedge clock or negedge reset)
begin
                if(~reset)              wdata0 <= 8'h00;
                else                    wdata0 <= btn2[0] ? wdata0+1'b1 : wdata0 ;
end
```

spi_task(spi_task.v)(2/4)

```
reg             [7:0]                   wdata1;
always @(posedge clock or negedge reset)
begin
                if(~reset)              wdata1 <= 8'h00;
                else                    wdata1 <= btn2[1] ? wdata1+1'b1 : wdata1 ;
end

reg             [7:0]                   wdata2;
always @(posedge clock or negedge reset)
begin
                if(~reset)              wdata2 <= 8'h00;
                else                    wdata2 <= btn2[2] ? wdata2+1'b1 : wdata2 ;
end

reg             [7:0]                   wdata3;
always @(posedge clock or negedge reset)
begin
                if(~reset)              wdata3 <= 8'h00;
                else                    wdata3 <= btn2[3] ? wdata3+1'b1 : wdata3 ;
end

// ---------------------------------------------
parameter       SLAVE_IDW = 8'h64;
parameter       SLAVE_IDR = 8'h65;
parameter       REG_ADDR1 = 8'h10;
parameter       REG_ADDR2 = 8'h11;
parameter       REG_ADDR3 = 8'h12;
parameter       REG_ADDR4 = 8'h13;
parameter    SPI_FREQ  = 10'd16;

// ---------------------------------------------
// 1) define state
reg             [2:0]                   m_state ;
parameter       M_IDLE   = 3'd0 ;
parameter       M_WREADY = 3'd1 ;
parameter       M_WRITE  = 3'd2 ;
parameter       M_RREADY = 3'd3 ;
parameter       M_READ   = 3'd4 ;

// ---------------------------------------------
// 2) state flag
wire            s_idle   = (m_state == M_IDLE   ) ? 1'b1 : 1'b0 ;
wire            s_wready = (m_state == M_WREADY ) ? 1'b1 : 1'b0 ;
wire            s_write  = (m_state == M_WRITE  ) ? 1'b1 : 1'b0 ;
wire            s_rready = (m_state == M_RREADY ) ? 1'b1 : 1'b0 ;
wire            s_read   = (m_state == M_READ   ) ? 1'b1 : 1'b0 ;

// ---------------------------------------------
// 3) code implementation
reg             [1:0]                   wready_cnt;
always @(posedge clock or negedge reset)
begin
                if(~reset)              wready_cnt <= 2'b0;
                else                    wready_cnt <= ~s_wready ? 2'b0 : wready_cnt+1'b1;
end
```

- U0 : spi_task(spi_task.v)

spi_task(spi_task.v)(3/4)

```
wire        [7:0]       reg_addr  = (btn_id==2'd0) ? REG_ADDR1 :
            (btn_id==2'd1) ? REG_ADDR2 :
            (btn_id==2'd2) ? REG_ADDR3 : REG_ADDR4 ;
reg         [7:0]       addr;
always @(posedge clock or negedge reset)
begin
            if(~reset)      addr <= 8'b0;
            else            addr <= (s_wready & (wready_cnt==2'b11)) ? reg_addr : addr ;
end

wire        [7:0]       reg_wdata = (btn_id==2'd0) ? wdata0 :
            (btn_id==2'd1) ? wdata1 :
            (btn_id==2'd2) ? wdata2 : wdata3 ;
reg         [7:0]       wdata;
always @(posedge clock or negedge reset)
begin
            if(~reset)      wdata <= 8'b0;
            else            wdata <= (s_wready & (wready_cnt==2'b11)) ? reg_wdata : wdata ;
end

reg                     start_w;
always @(posedge clock or negedge reset)
begin
            if(~reset)      start_w <= 1'b0;
            else            start_w <= (s_wready & (wready_cnt==2'b11)) ? 1'b1 : 1'b0 ;
end

reg         [3:0]       write_cnt;
always @(posedge clock or negedge reset)
begin
            if(~reset)      write_cnt <= 4'b0;
            else            write_cnt <= ~s_write ? 4'b0 :
                    (write_cnt==4'd15) ? 4'd15 : write_cnt+1'b1;
end

reg         [1:0]       rready_cnt;
always @(posedge clock or negedge reset)
begin
            if(~reset)      rready_cnt <= 2'b0;
            else            rready_cnt <= ~s_rready ? 2'b0 : rready_cnt+1'b1;
end

reg                     start_r;
always @(posedge clock or negedge reset)
begin
            if(~reset)      start_r <= 1'b0;
            else            start_r <= (s_rready & (rready_cnt==2'b11)) ? 1'b1 : 1'b0 ;
end
```

spi_task(spi_task.v)(4/4)

```
reg         [3:0]       read_cnt;
always @(posedge clock or negedge reset)
begin
            if(~reset)      read_cnt <= 4'b0;
            else            read_cnt <= ~s_read ? 4'b0 :
                    (read_cnt==4'd15) ? 4'd15 : read_cnt+1'b1;
end

wire        [7:0]       rdata  ;
wire                    ss     ;
wire                    sck    ;
wire                    mosi   ;
wire                    miso   ;
wire                    done_mst;
spi_master  spi_master_u1(.reset   (reset   ),.clock   (clock   ),.freq    (SPI_FREQ ),
            .start_w  (start_w  ), .start_r  (start_r  ),     .addr    (addr     ),
            .wdata   (wdata    ),     .rdata   (rdata    ),     .done    (done_mst ),
            .ss      (ss       ),     .sck     (sck      ),     .mosi    (mosi     ),
            .miso    (miso     )
);

spi_slave   spi_slave_u1(.reset    (reset   ), .clock    (clock   ), .ss      (ss      ),
            .sck     (sck      ), .mosi    (mosi    ), .miso    (miso    )
);

reg         [7:0]       led;
always @(posedge clock or negedge reset)
begin
            if(~reset)      led <= 4'b0;
            else            led <= (s_read & (read_cnt==4'd15) & done_mst) ? rdata : led;
end

// ---------------------------------------------
// 4) state transition
always @(posedge clock or negedge reset)
begin
            if(~reset)      m_state <= 3'b0;
            else            m_state <= (s_idle   & (btn2 != 4'b0 )              ) ? M_WREADY :
                    (s_wready & (wready_cnt==2'b11)          ) ? M_WRITE  :
                    (s_write  & (write_cnt==4'd15) & done_mst ) ? M_RREADY :
                    (s_rready & (rready_cnt==2'b11)          ) ? M_READ   :
                    (s_read   & (read_cnt==4'd15) & done_mst  ) ? M_IDLE   :
                     m_state ;
end

endmodule
```

Create and Package New IP :SPI

➢ top_fndcontrol2

top_fndcontrol2(1/2)

```
`timescale 1ns / 1ps
module top_fndcontrol2(
    input clk,
    input reset,
input mode,
    input [3:0] btn,
output  [3:0] fndsel,
    output  [6:0] fnd,
    output [7:0] led_out
);

wire [7:0] led;
wire  w_clkout;
wire [1:0] w_counter;
wire [3:0] w_datafnd;
wire [3:0] w_outa, w_outb, w_outc, w_outd;
wire [3:0] leda;
wire  [3:0] ledb;
assign w_outa = {3'b000,btn[0]};
assign w_outb = {2'b00,btn[1],1'b0};
assign w_outc = {1'b0,btn[2],2'b00};
assign w_outd = {btn[3],3'b000};
assign leda = led[3:0];
assign ledb = led[7:4];
assign led_out = {ledb, leda};
 spi_task U0(
                 .reset(reset)   ,
                 .clock(clk)   ,
                 .btn (btn)    ,
                 .led(led)
    );
```

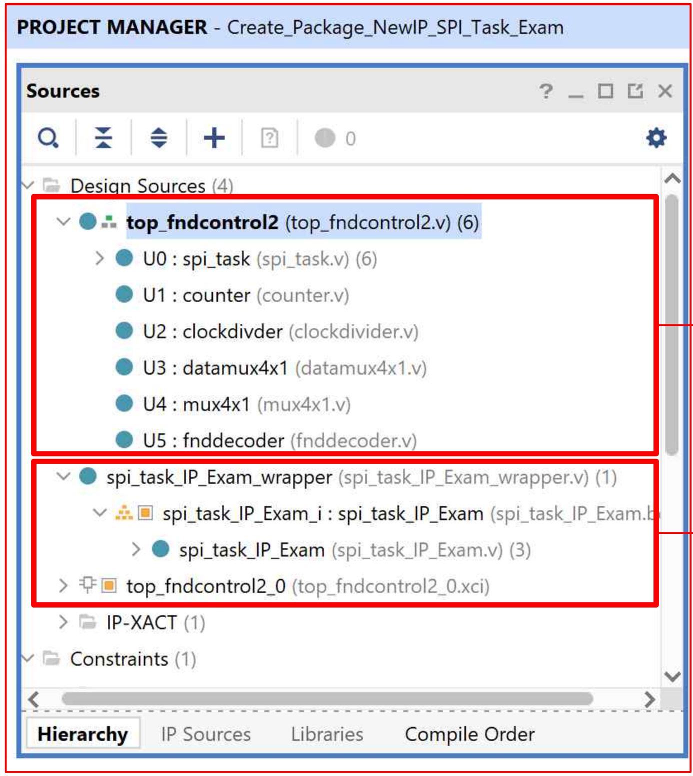
top_fndcontrol2(2/2)

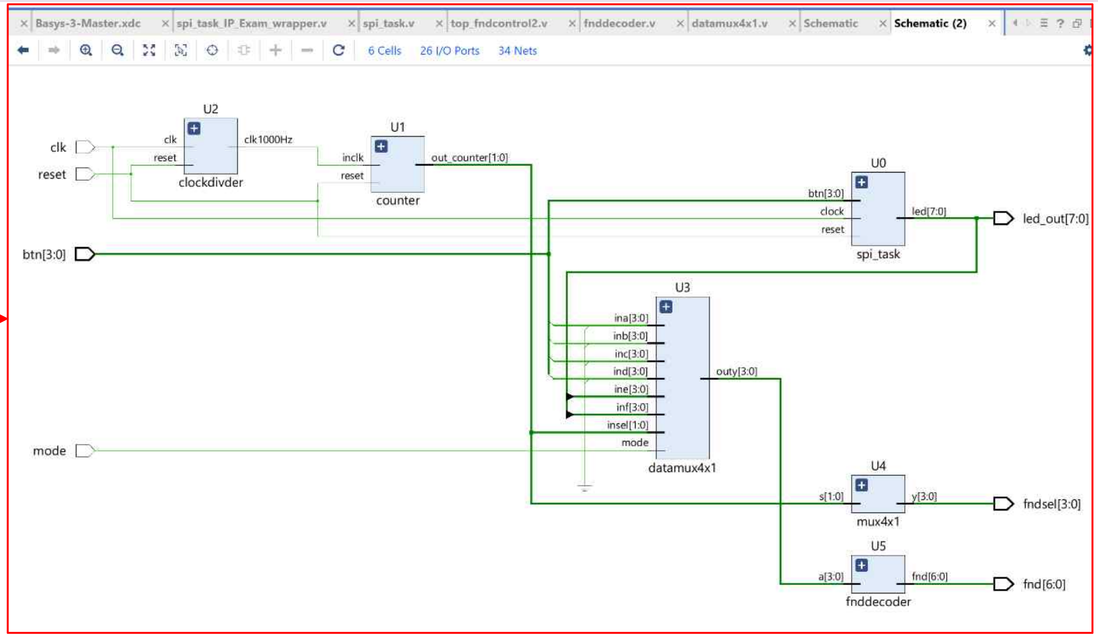
```
  counter U1 (
       .inclk(w_clkout),
       .reset(reset),
       .out_counter(w_counter)
   );
  clockdivder U2 (
       .clk(clk),
       .reset(reset),
       .clk1000Hz(w_clkout)
   );
      datamux4x1 U3 (
       .mode(mode),
       .ina  (w_outa),
       .inb  (w_outb),
       .inc  (w_outc),
       .ind  (w_outd),
       .ine  (leda),
       .inf  (ledb),
       .insel(w_counter),
       .outy  (w_datafnd)
   );
   mux4x1 U4 (
.s(w_counter),
       .y(fndsel)
   );
   fnddecoder U5 (
.reset(reset),
       .a(w_datafnd),
.fnd(fnd)
   );
endmodule
```

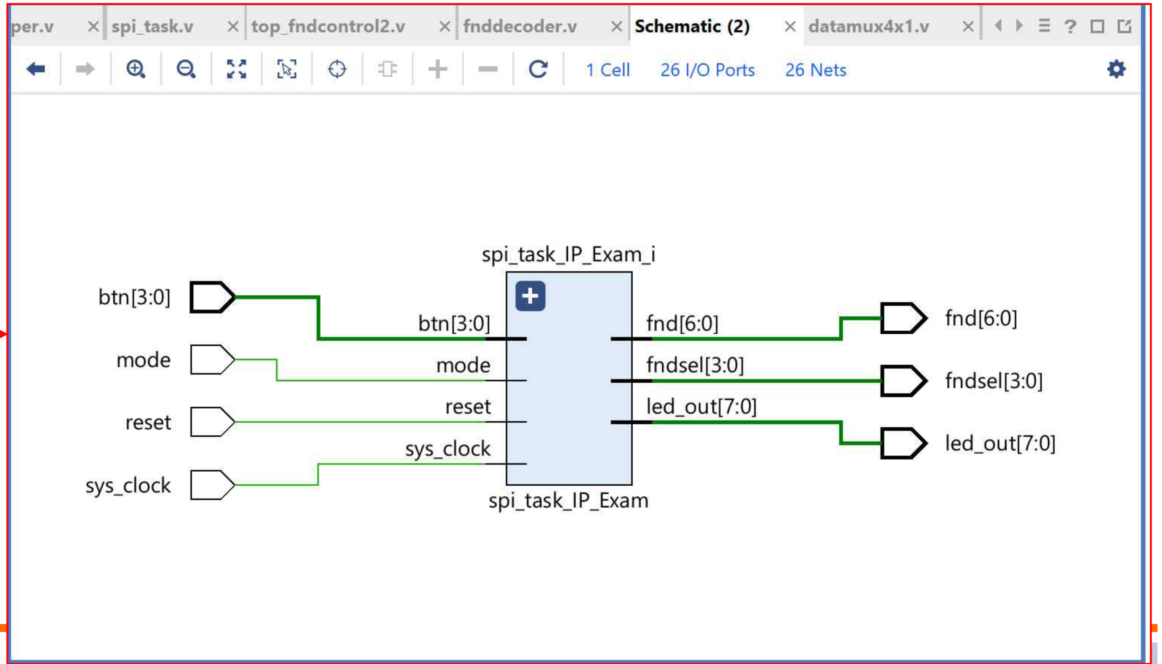
Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

- ***spi_task*** and ***top_fndcontrol2***
  - ***Design Source → Create or Add***

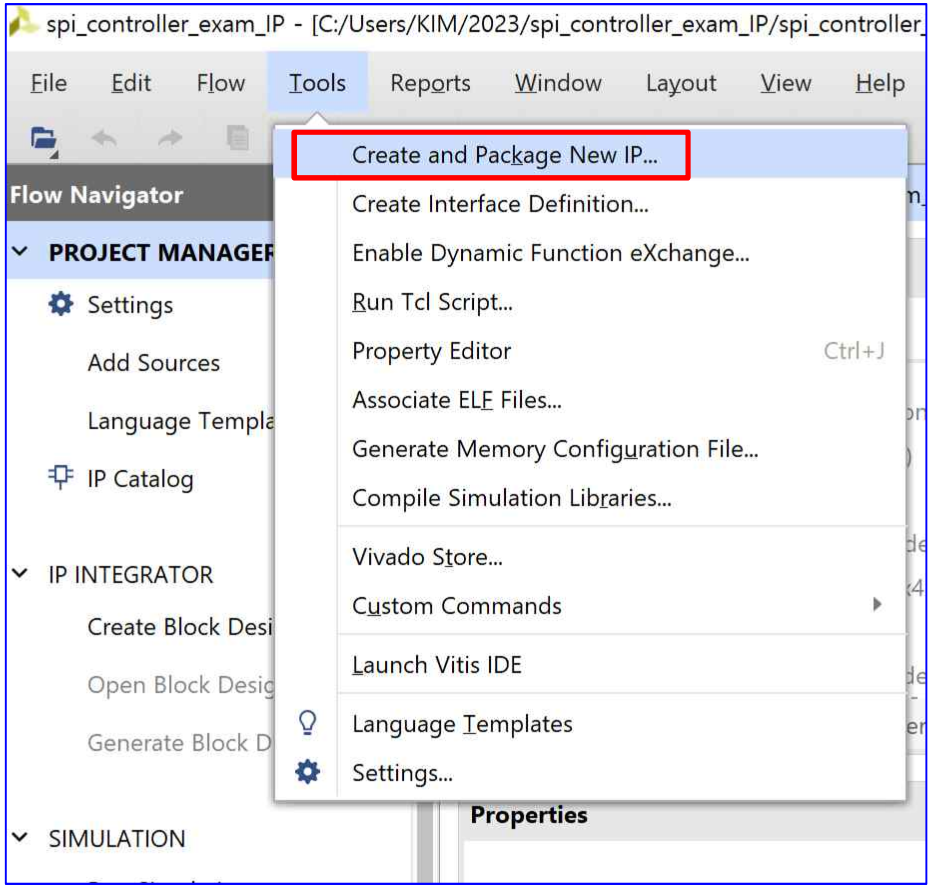






Create and Package New IP :SPI

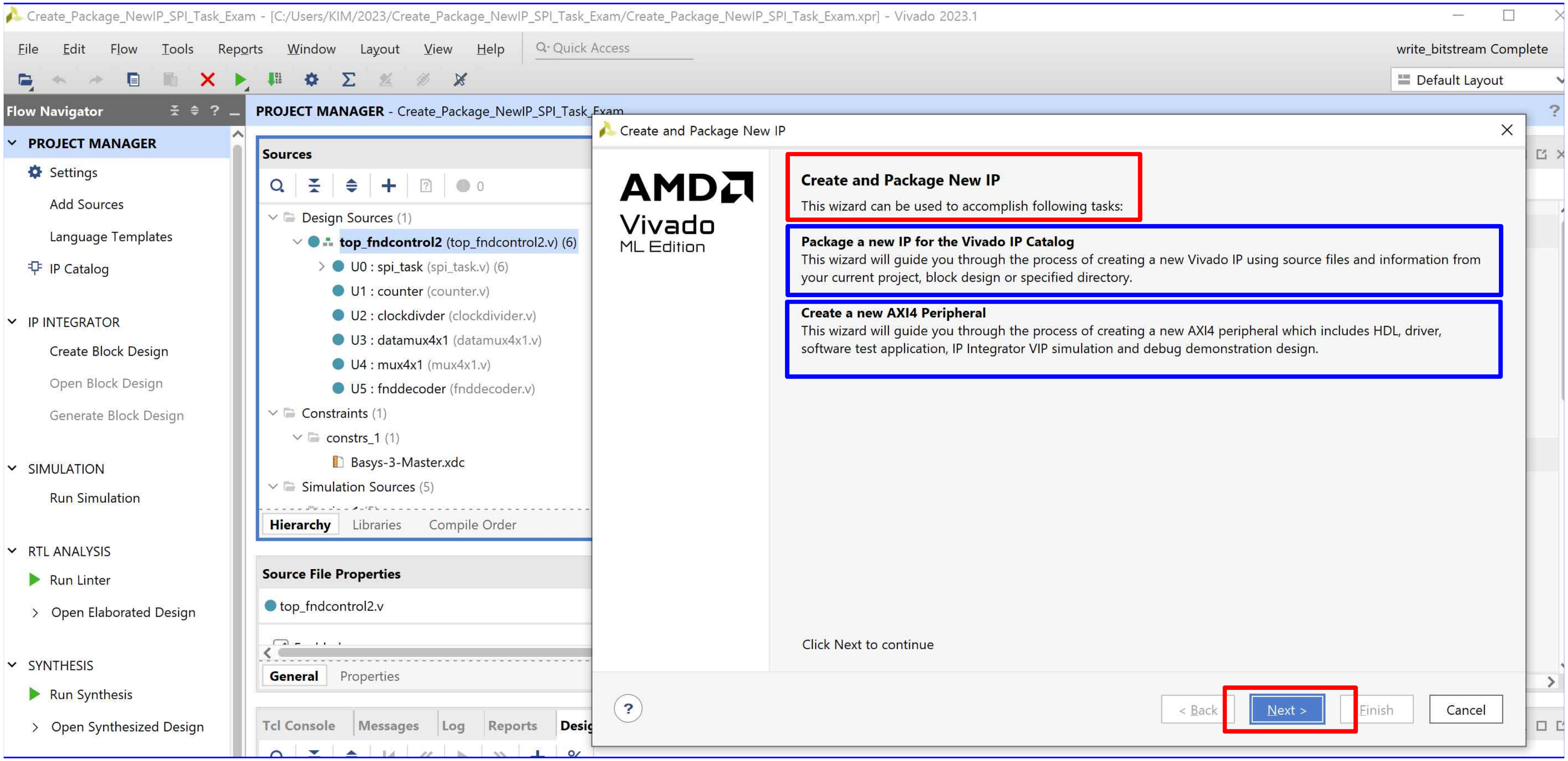
➤



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

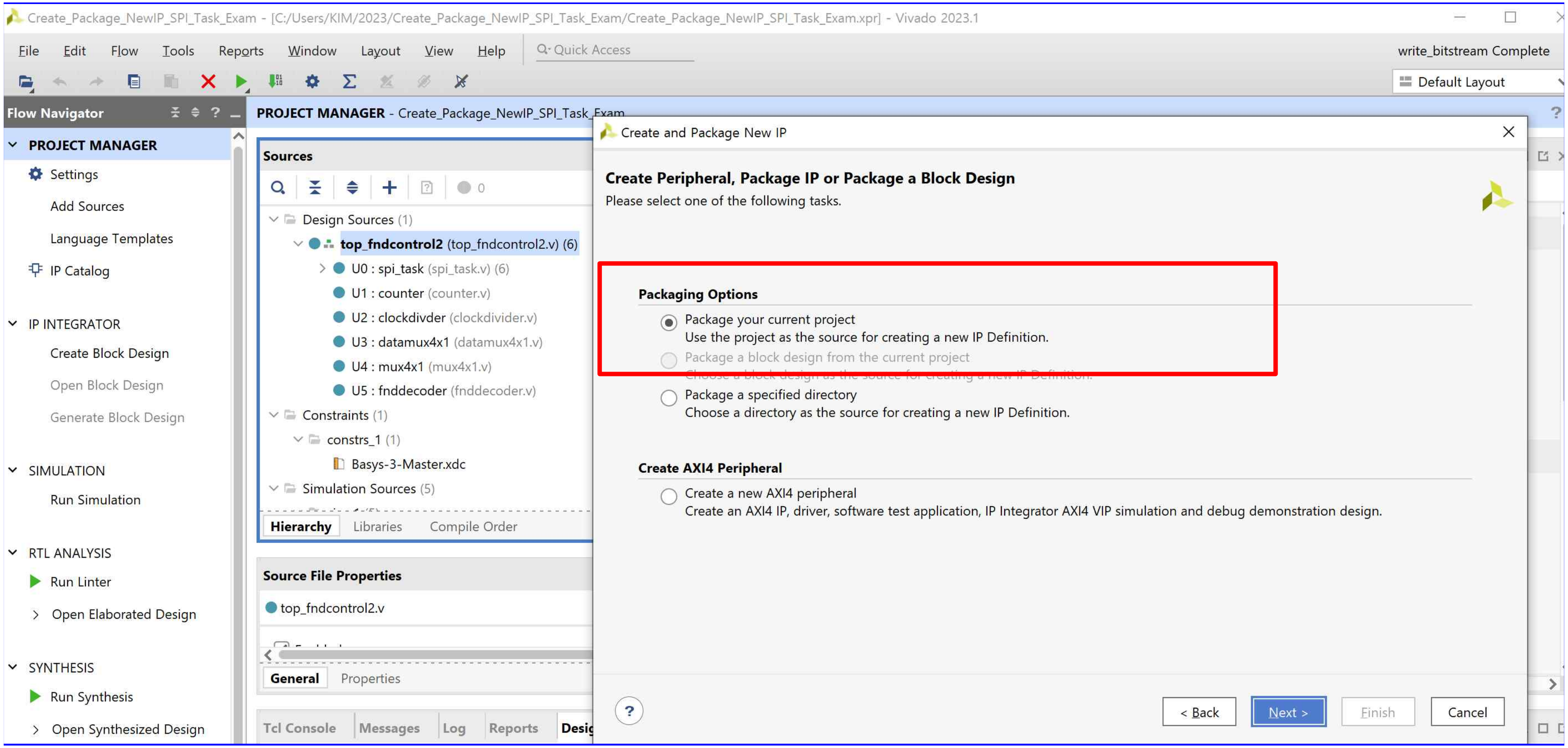
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

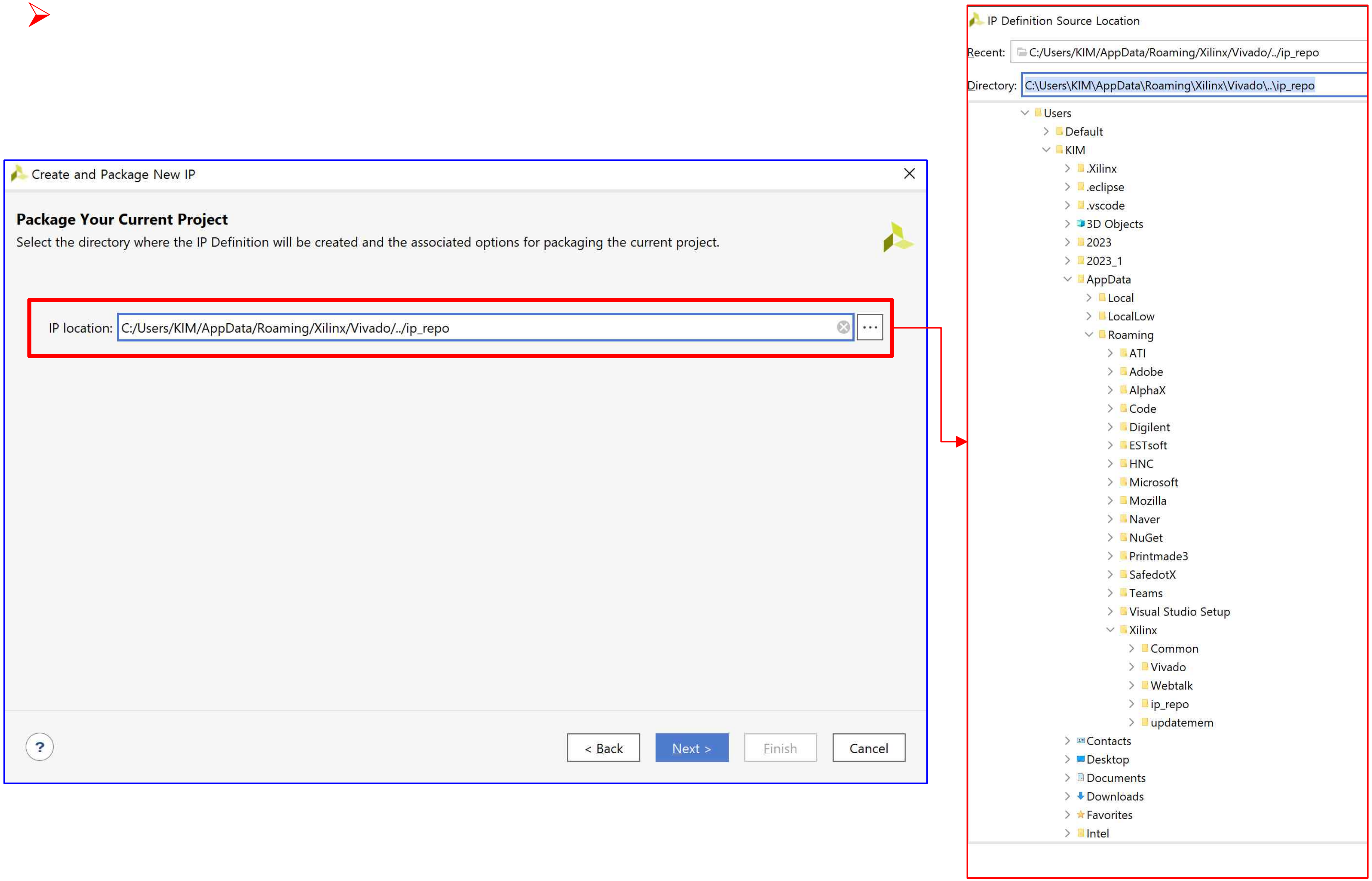
➢



Create and Package New IP :SPI

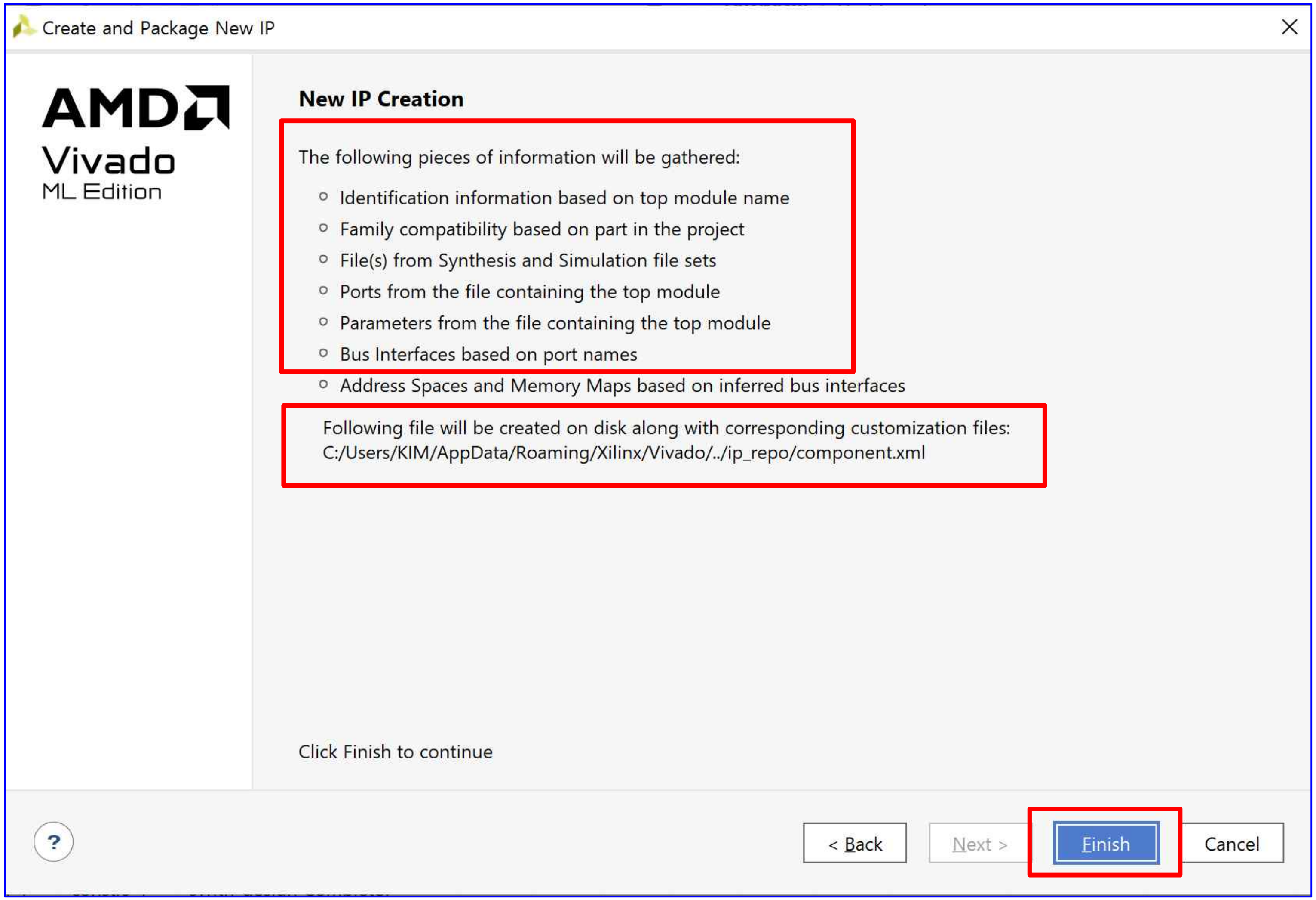
Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢



Create and Package New IP :SPI

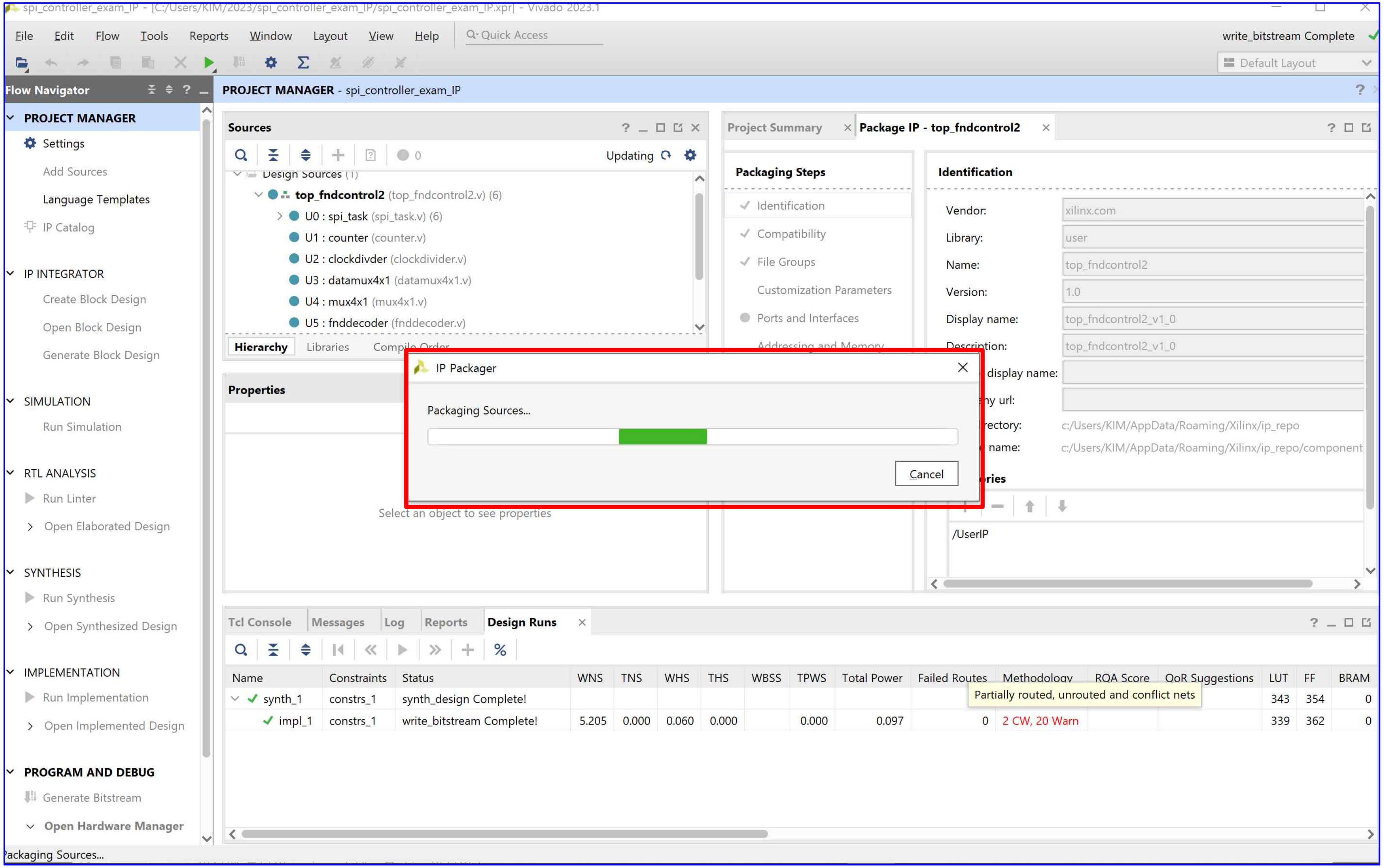
Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

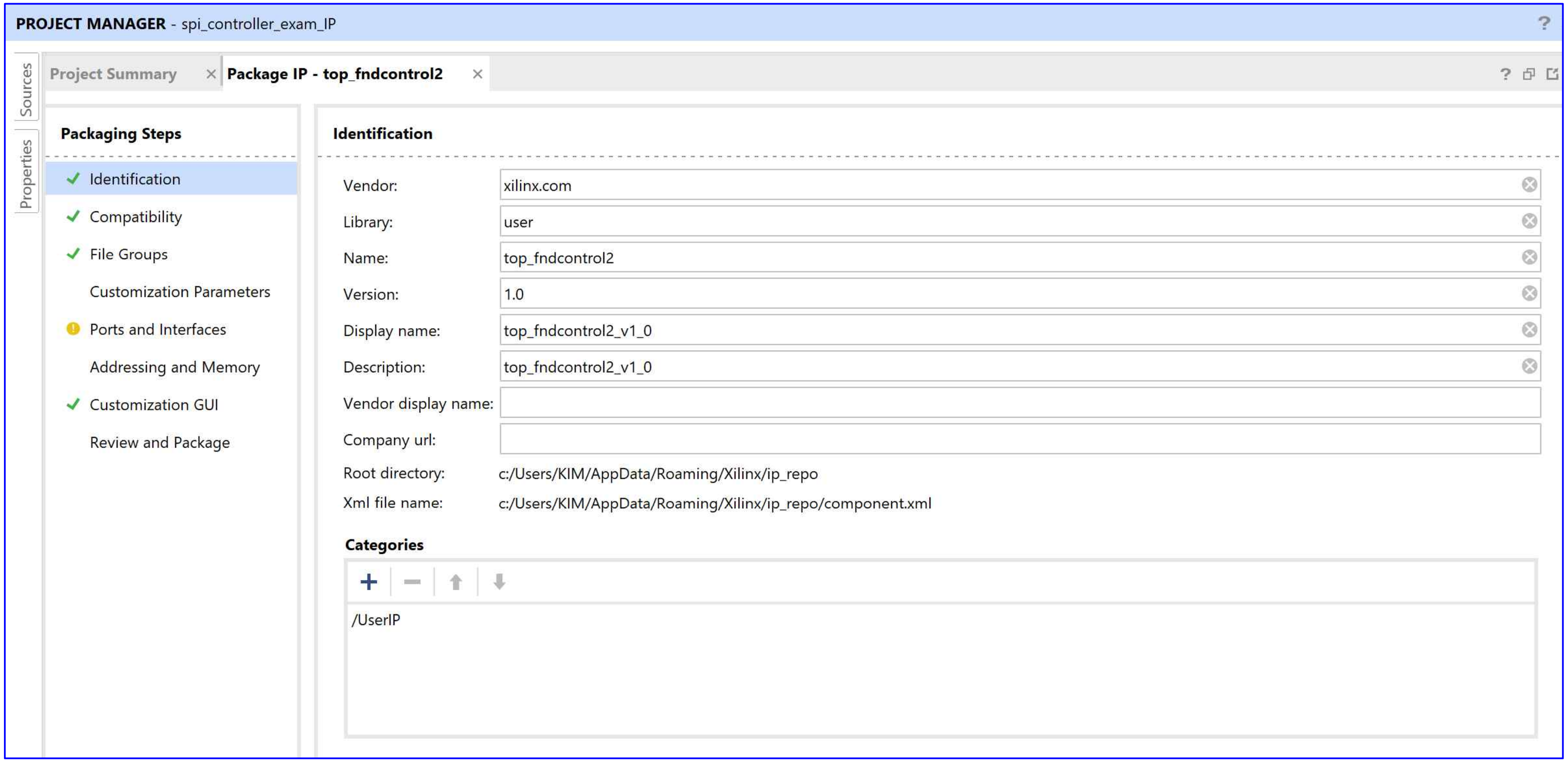
Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢



Create and Package New IP :SPI

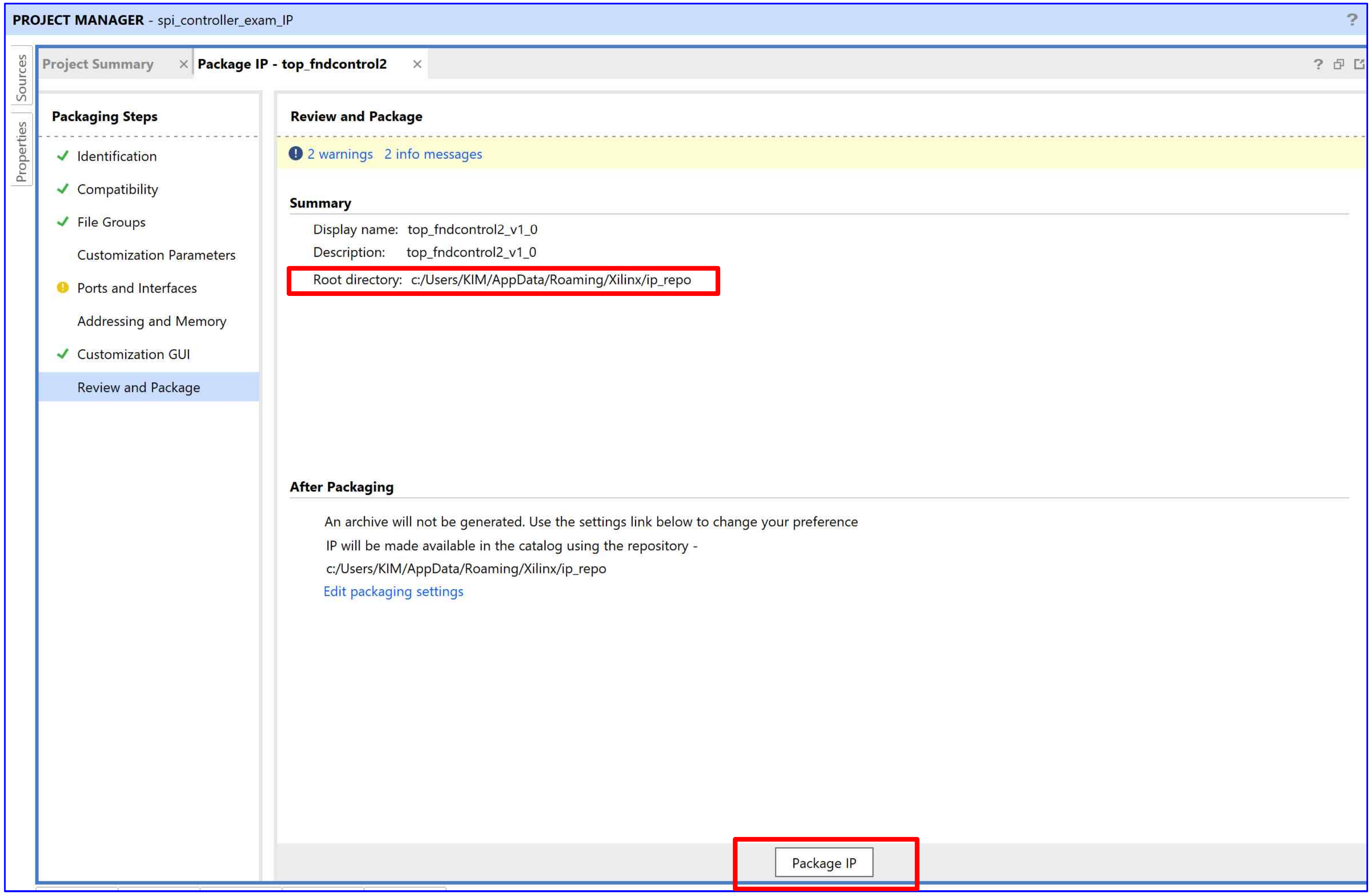
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

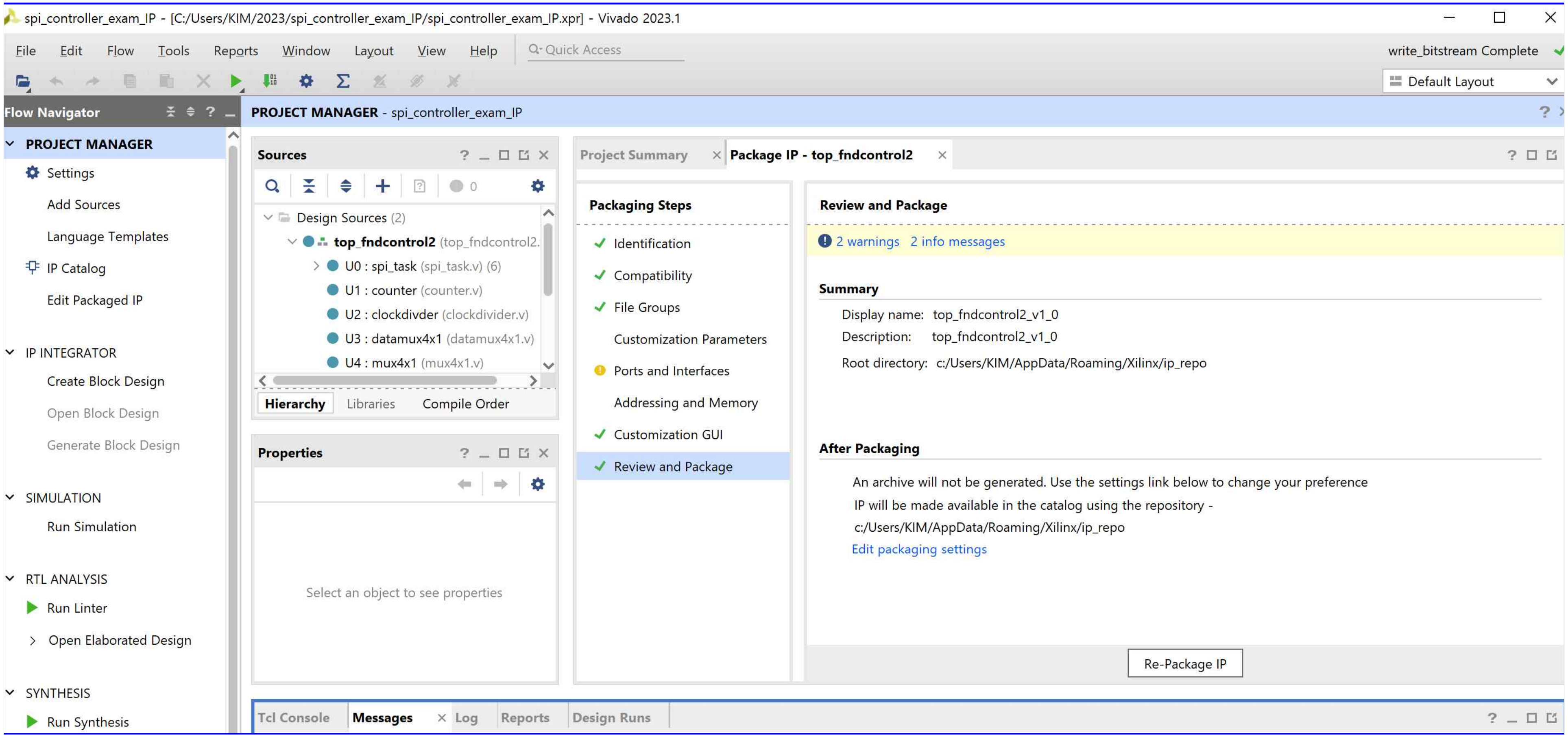
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

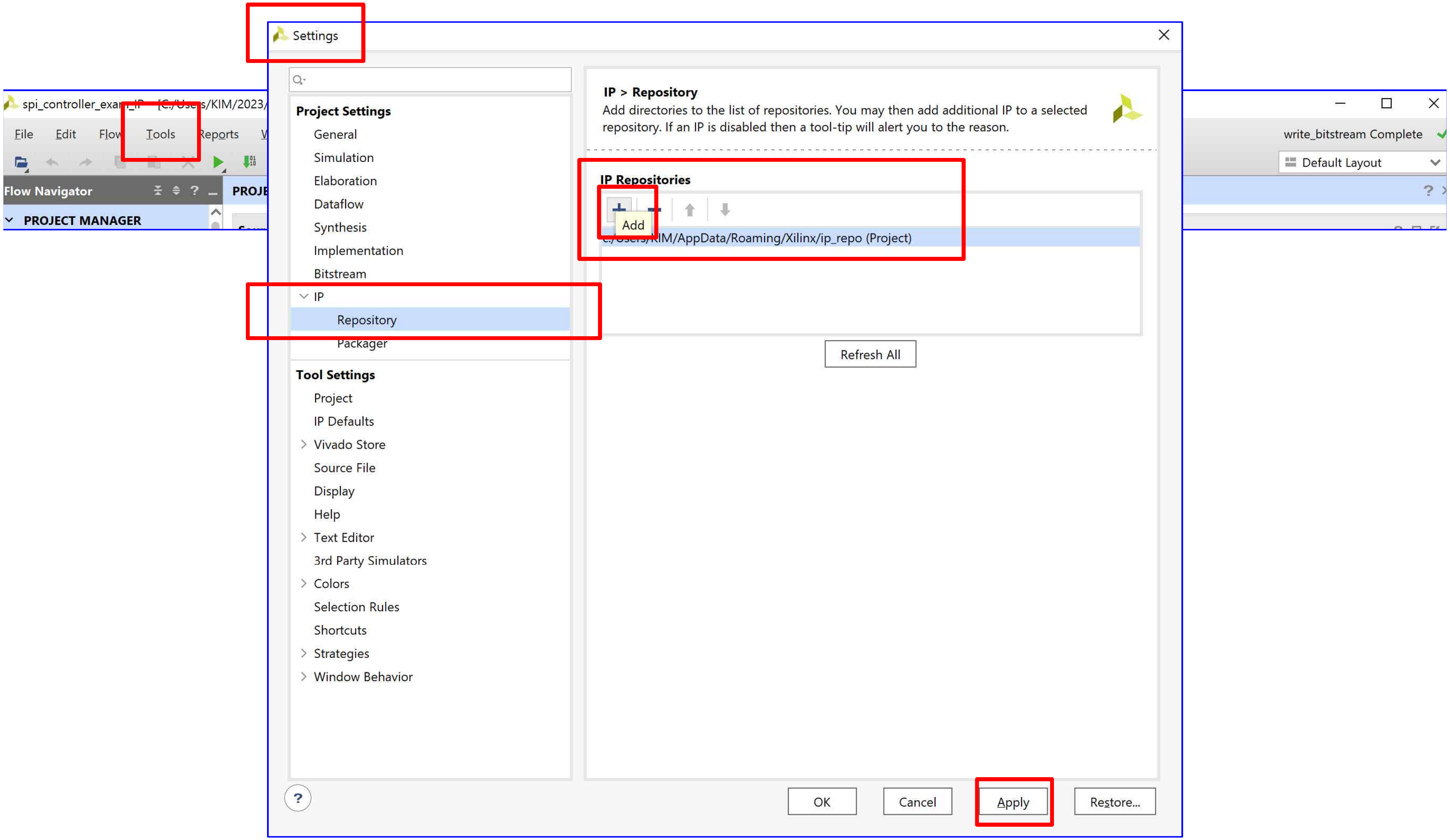
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

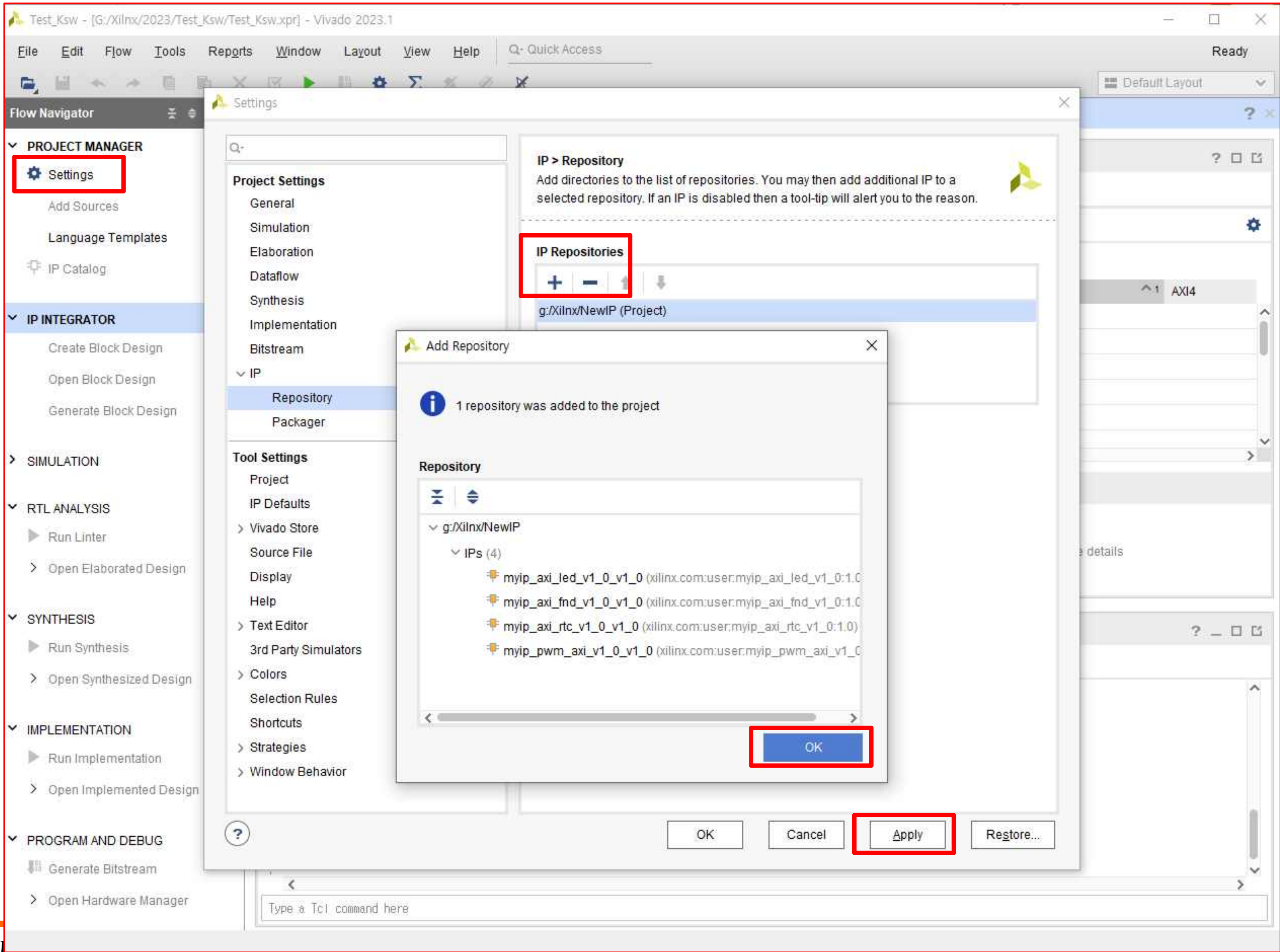
- ➢ ***IP ➔ ADD*** : *Tools → Setting → IP → Repository → ADD*



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

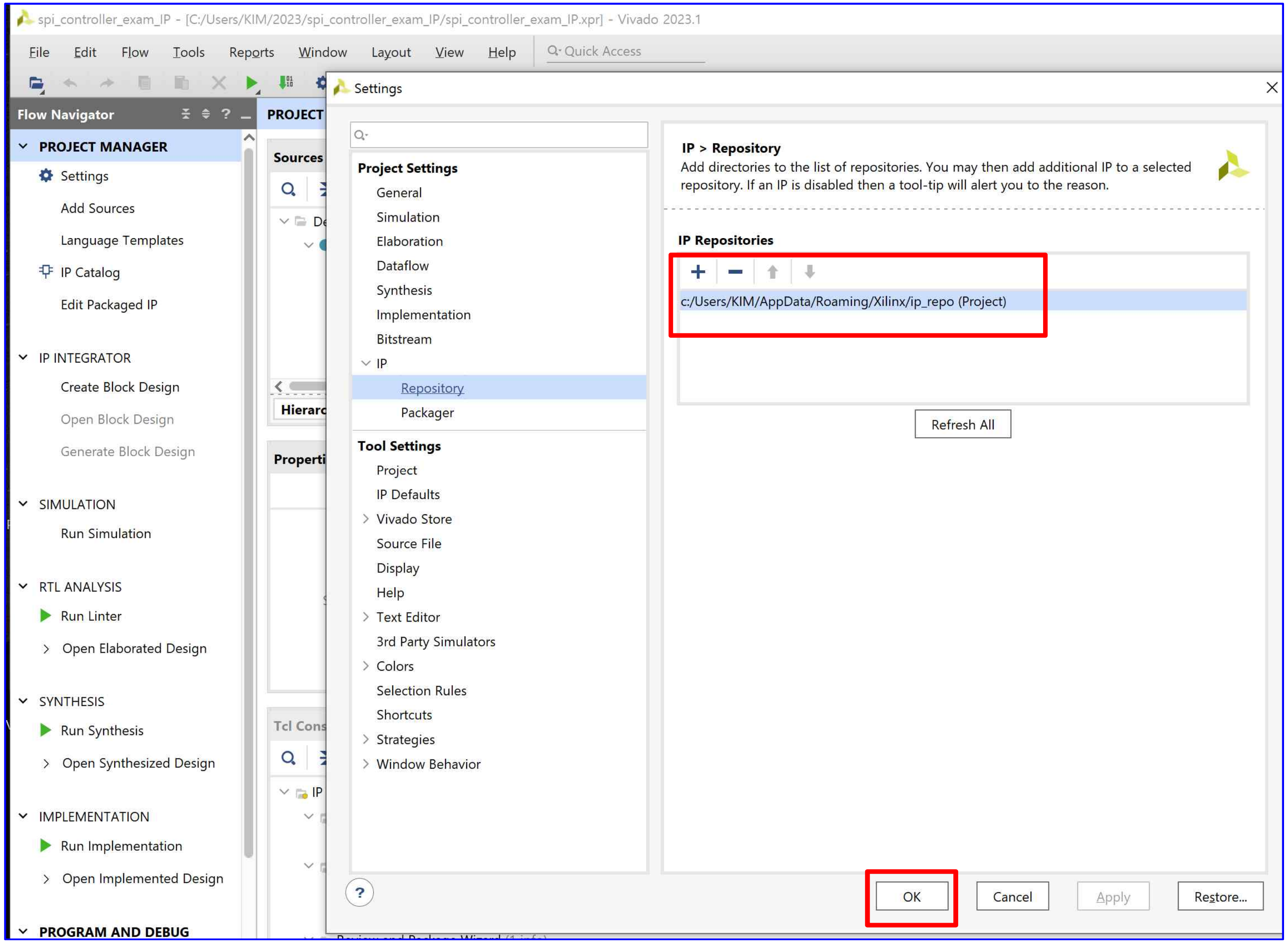
➢ IP ➔ ADD : Tools → Setting → IP → Repository → ADD



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

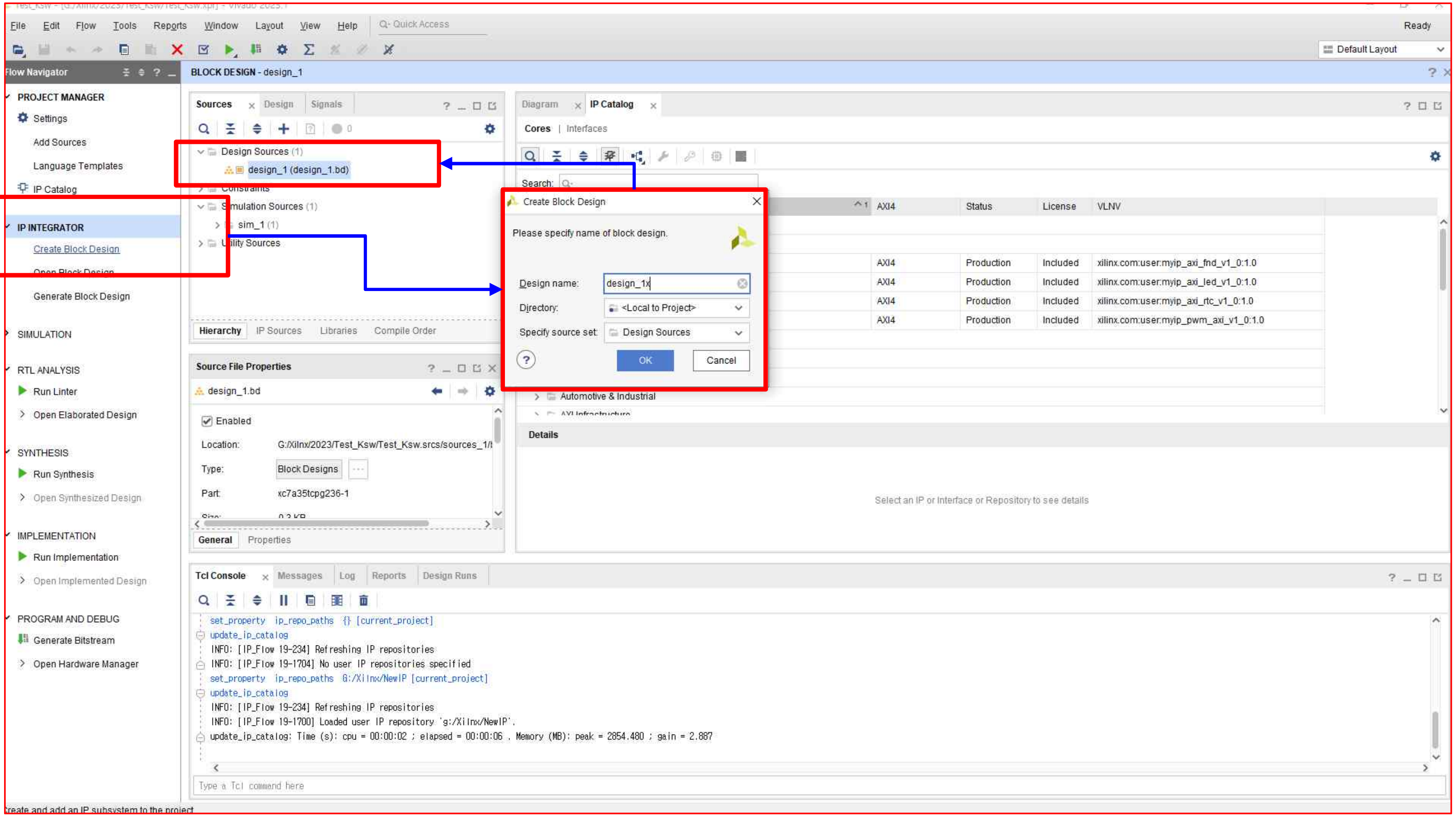
➢ ***IP → ADD*** : *Tools → Setting → IP → Repository → ADD*



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

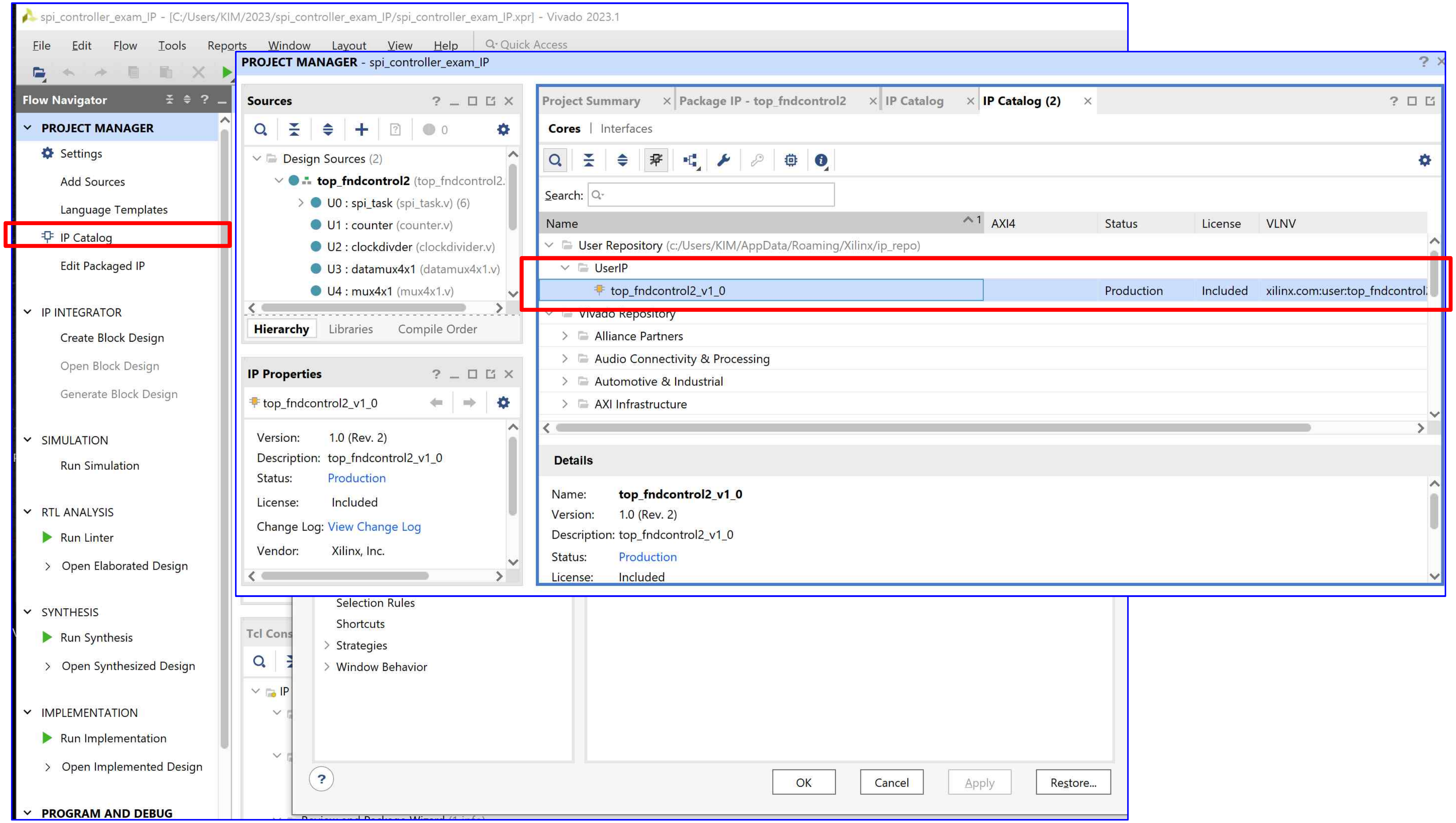
➢ ***IP INTEGRATOR ➔ Create Block Design***



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

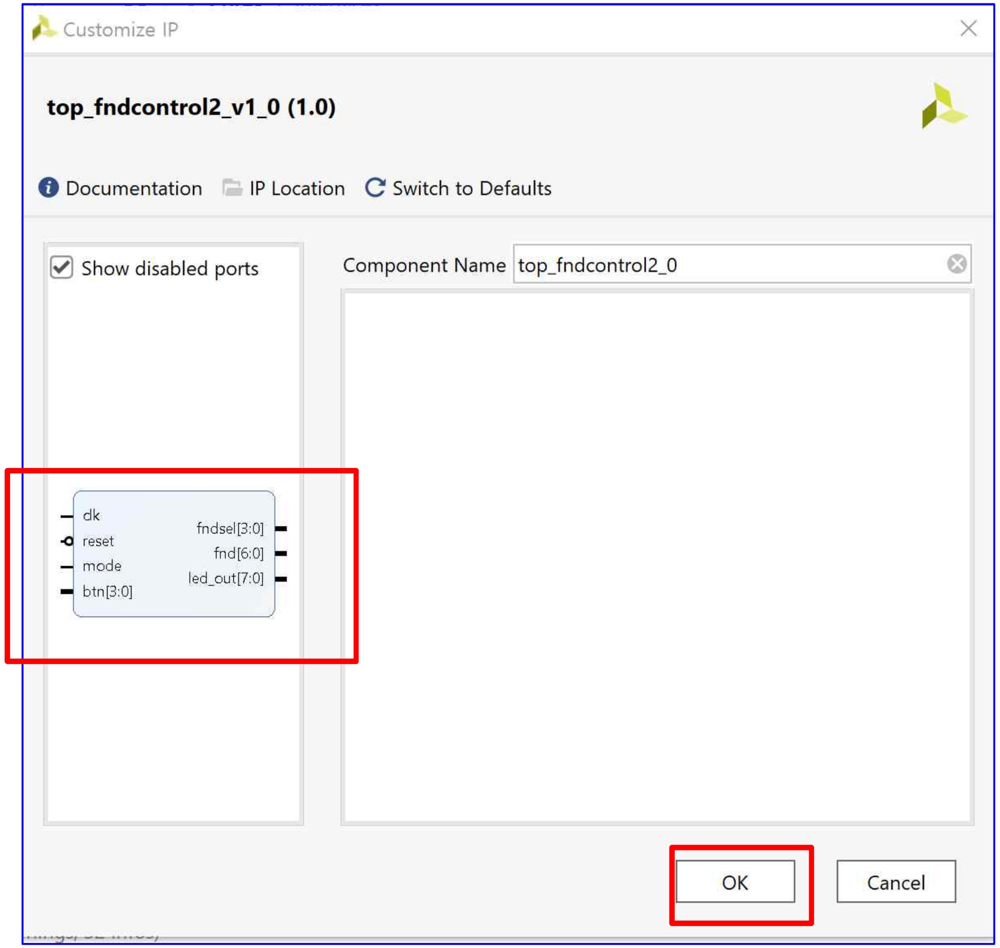
- ***IP INTEGATOR ➔ Create Block Design***
  - ***Usier IP ➔ …***

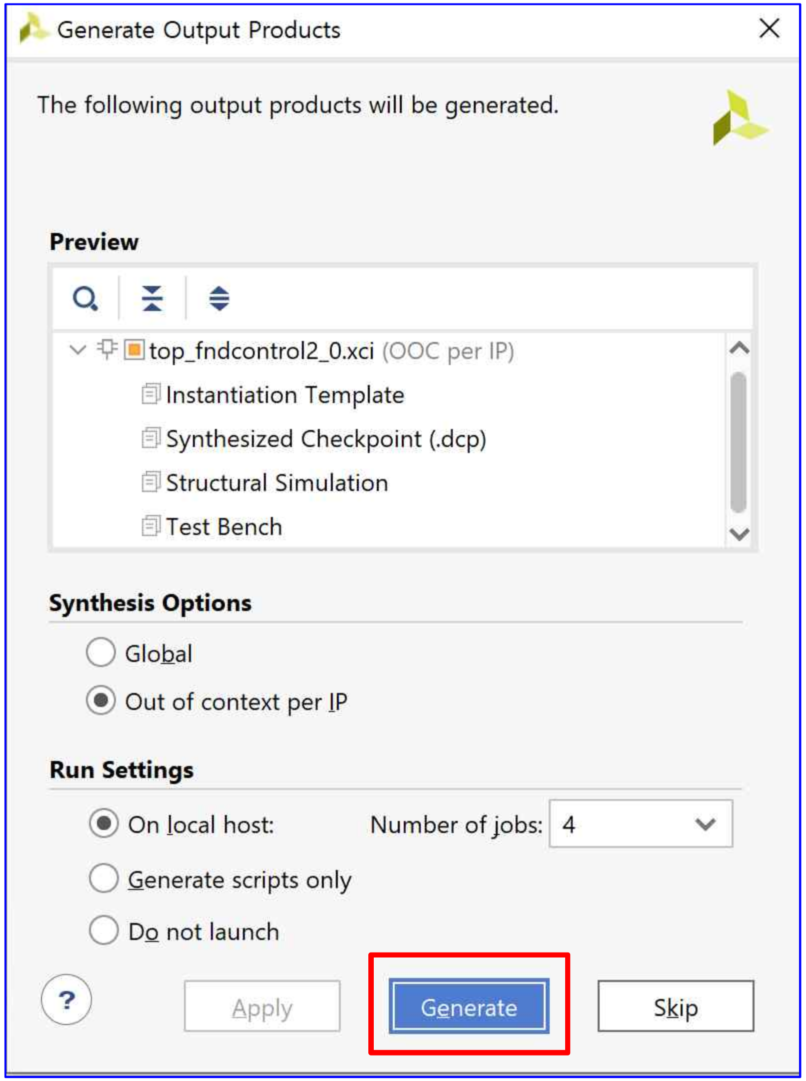


Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢

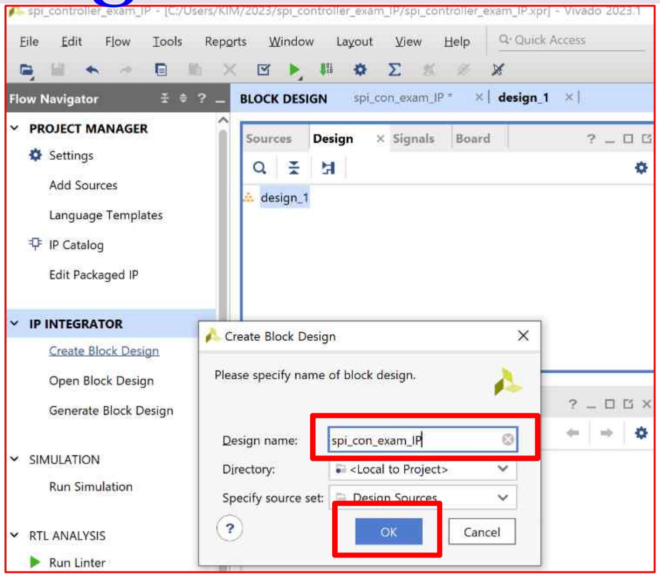


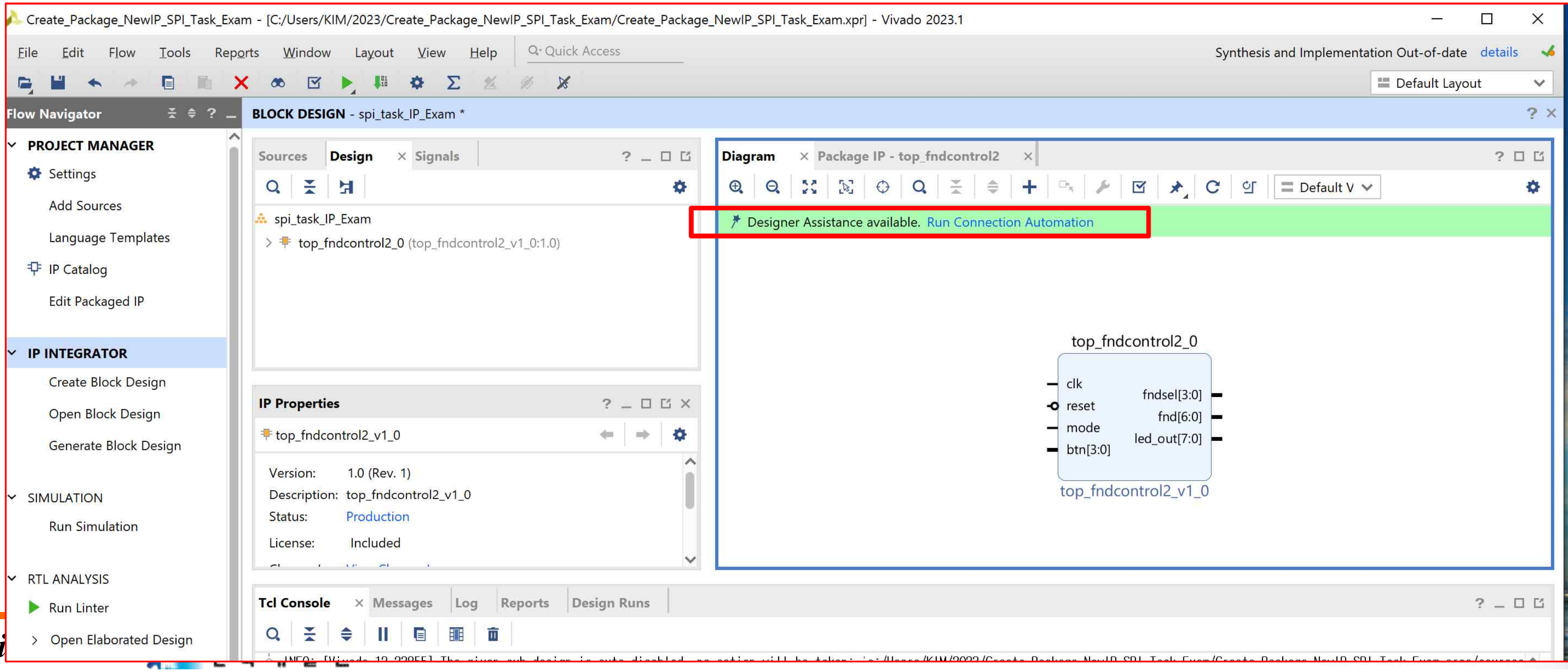


Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

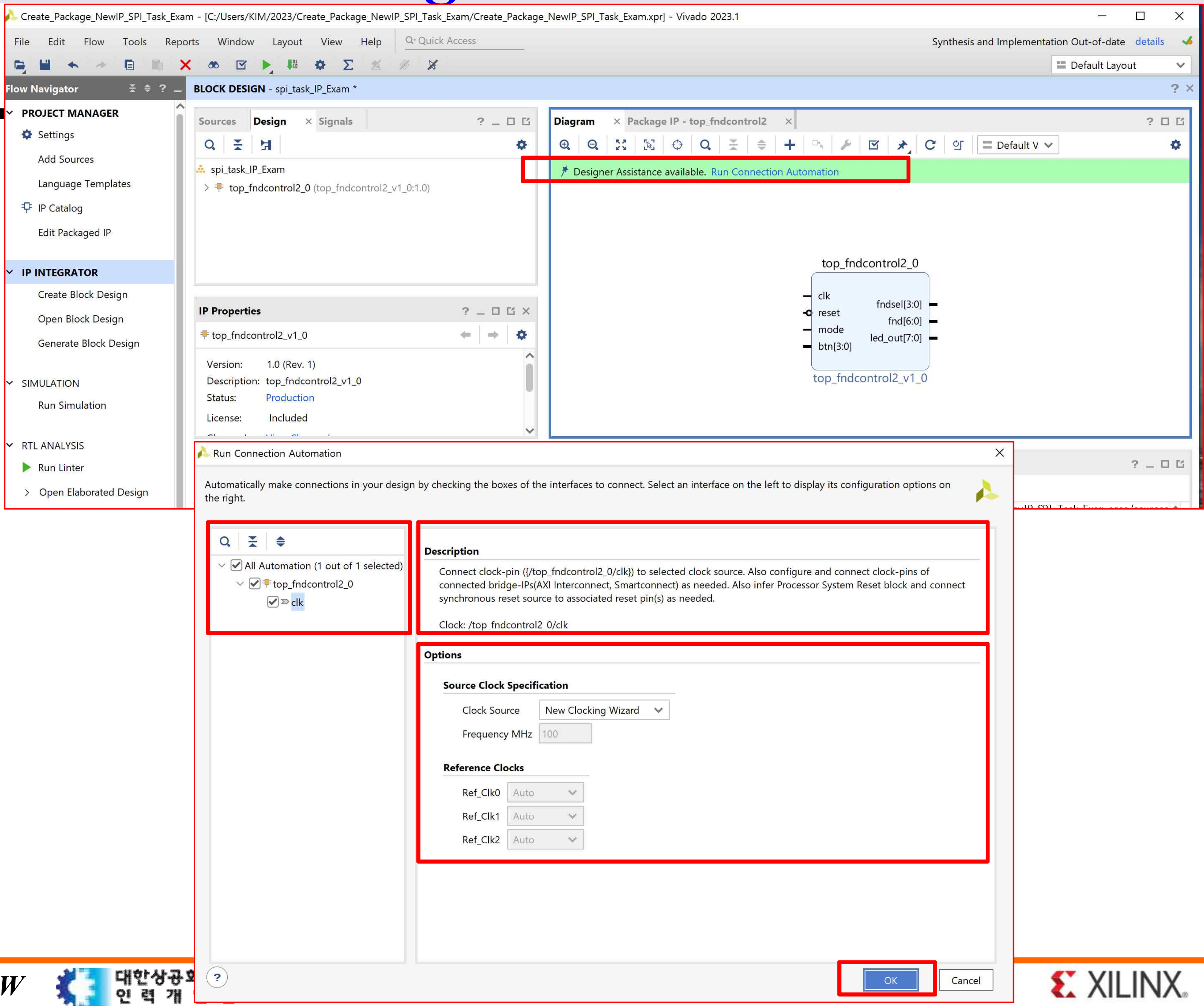
➤





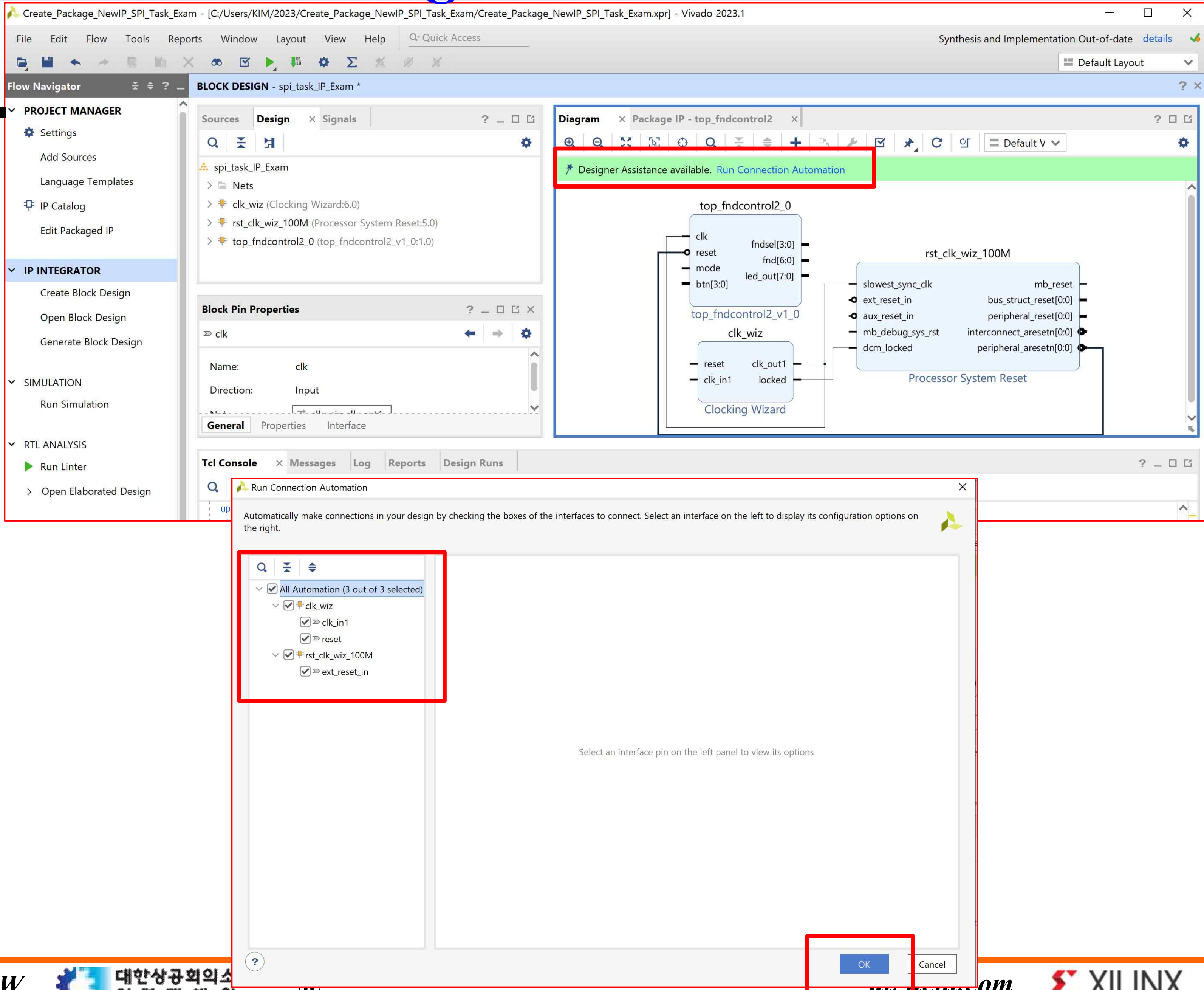
Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

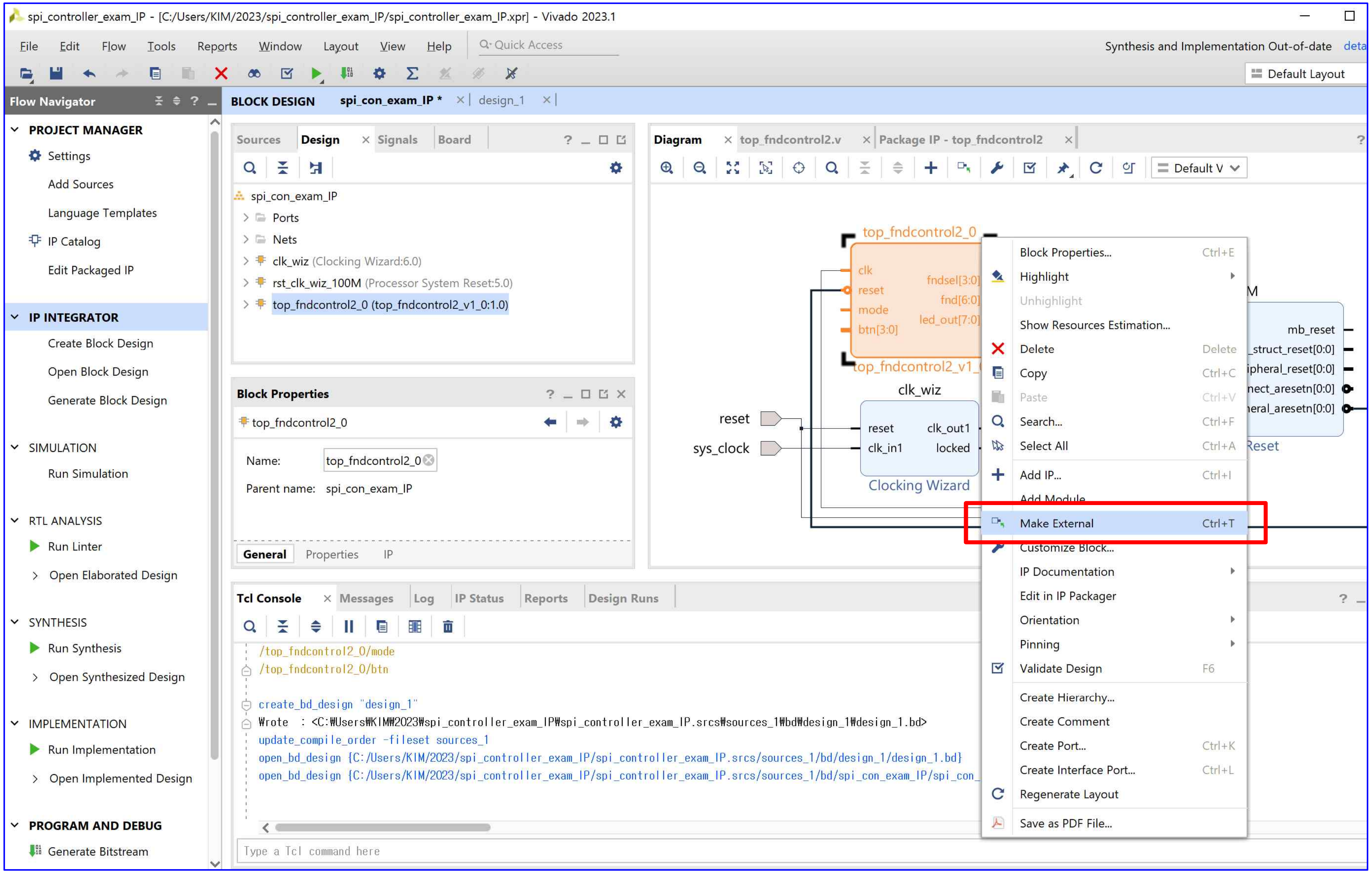
Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤



➤

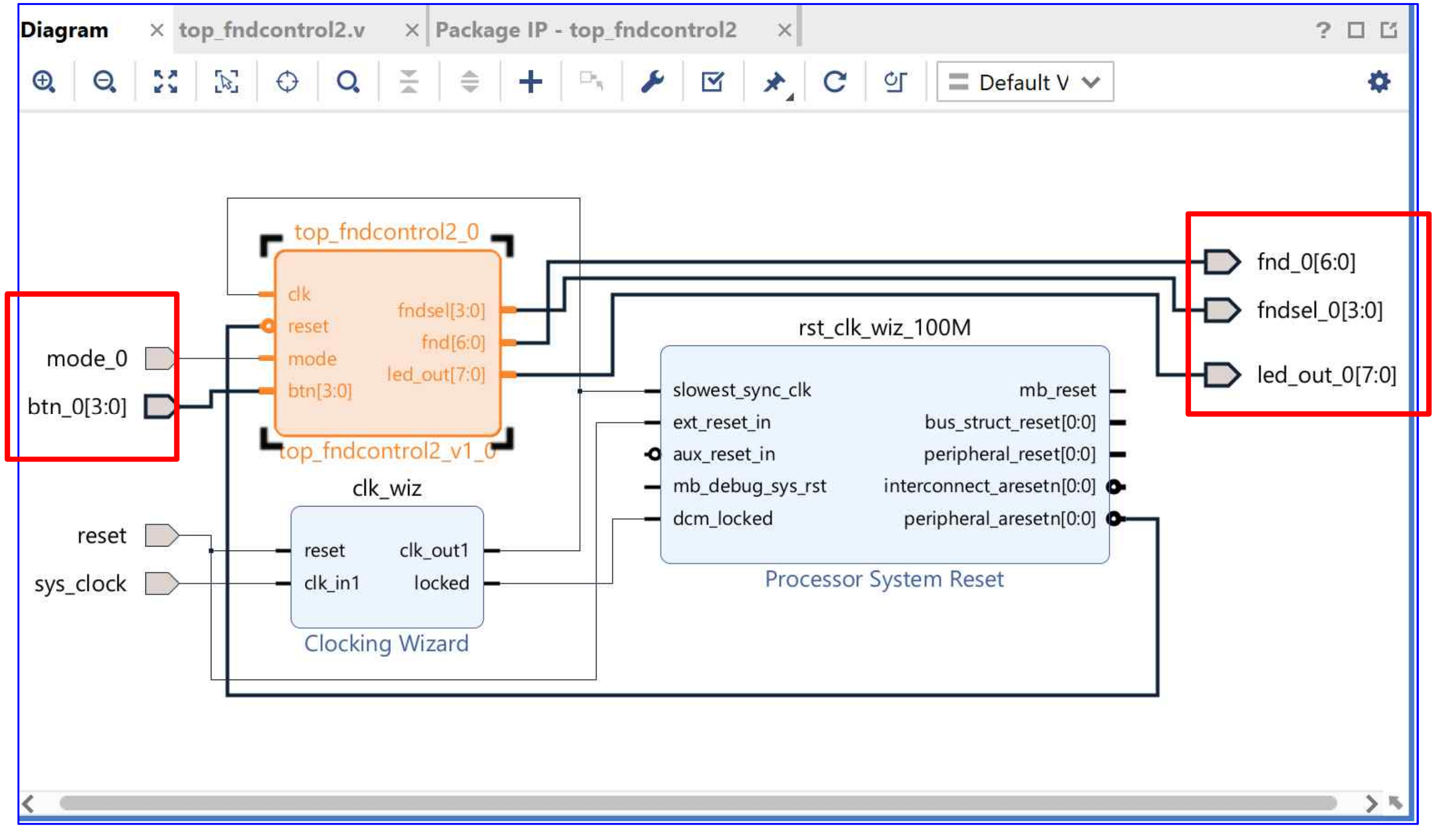
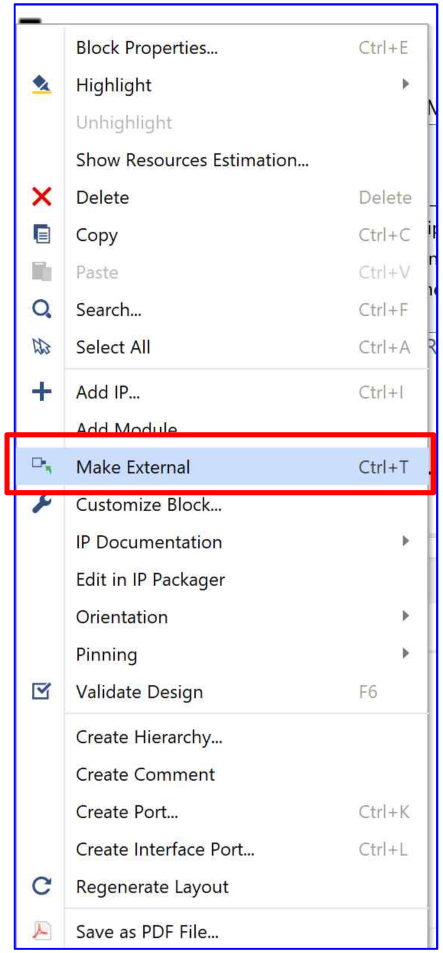


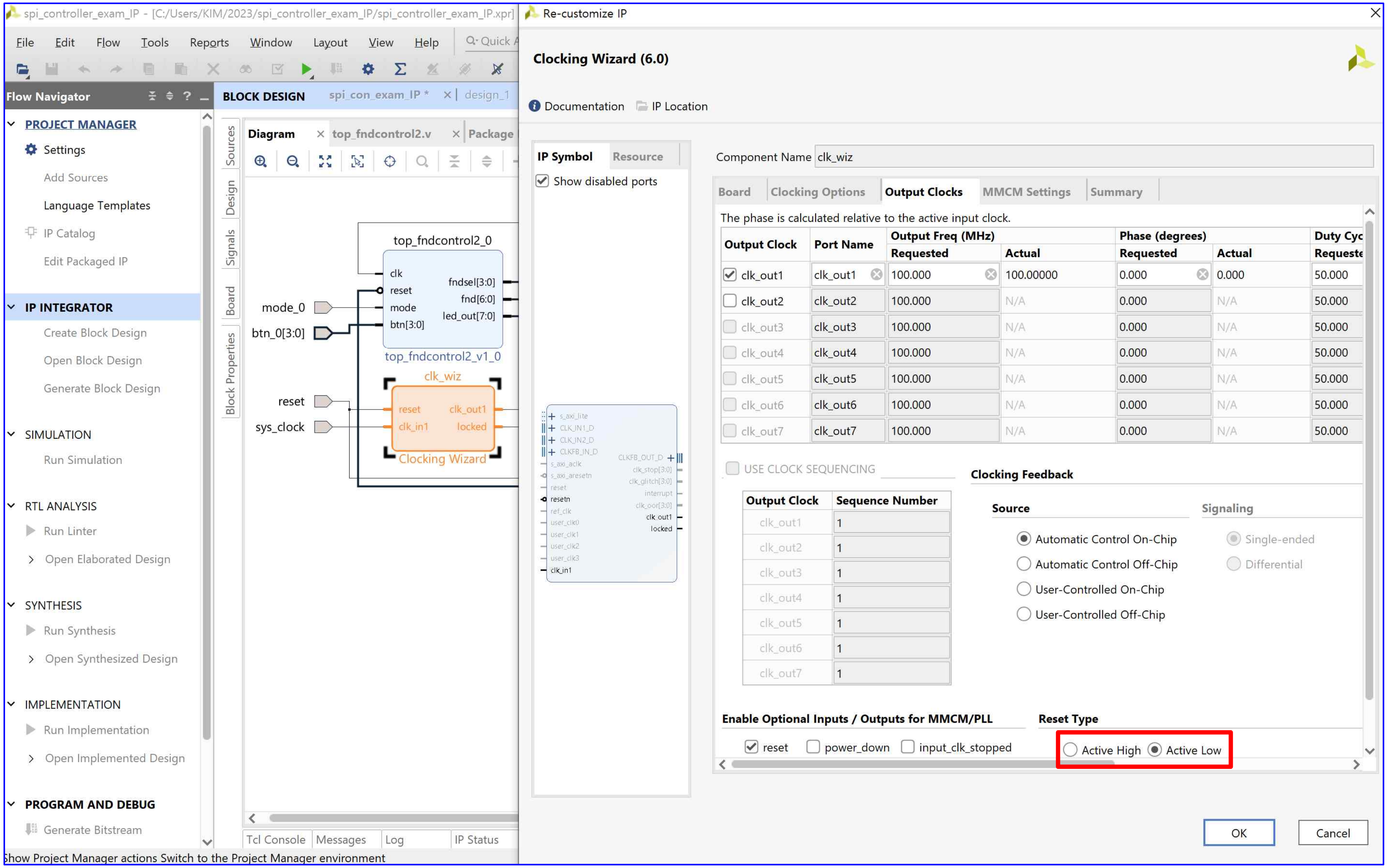
Diagram
top_fndcontrol2.v
Package IP - top_fndcontrol2
Default V
top_fndcontrol2_0
clk
reset
mode
btn[3:0]
fndsel[3:0]
fnd[6:0]
led_out[7:0]
top_fndcontrol2_v1_0
mode_0
btn_0[3:0]
fnd_0[6:0]
fndsel_0[3:0]
led_out_0[7:0]
rst_clk_wiz_100M
slowest_sync_clk
ext_reset_in
aux_reset_in
mb_debug_sys_rst
dcm_locked
mb_reset
bus_struct_reset[0:0]
peripheral_reset[0:0]
interconnect_aresetn[0:0]
peripheral_aresetn[0:0]
Processor System Reset
clk_wiz
reset
clk_in1
clk_out1
locked
Clocking Wizard
reset
sys_clock


Block Properties... Ctrl+E
Highlight
Unhighlight
Show Resources Estimation...
Delete Delete
Copy Ctrl+C
Paste Ctrl+V
Search... Ctrl+F
Select All Ctrl+A
Add IP... Ctrl+I
Add Module...
Make External Ctrl+T
Customize Block...
IP Documentation
Edit in IP Packager
Orientation
Pinning
Validate Design F6
Create Hierarchy...
Create Comment
Create Port... Ctrl+K
Create Interface Port... Ctrl+L
Regenerate Layout
Save as PDF File...

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

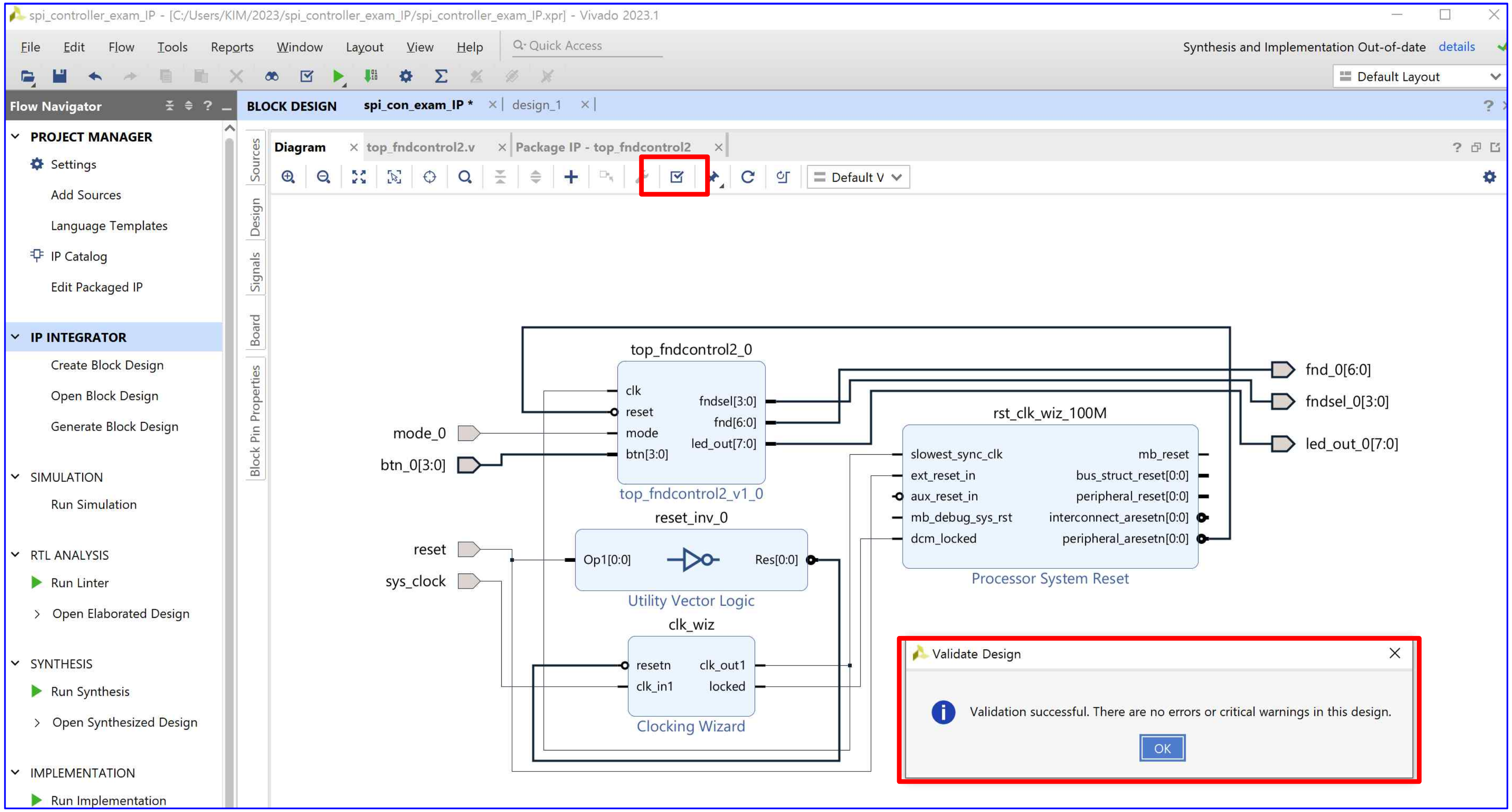
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

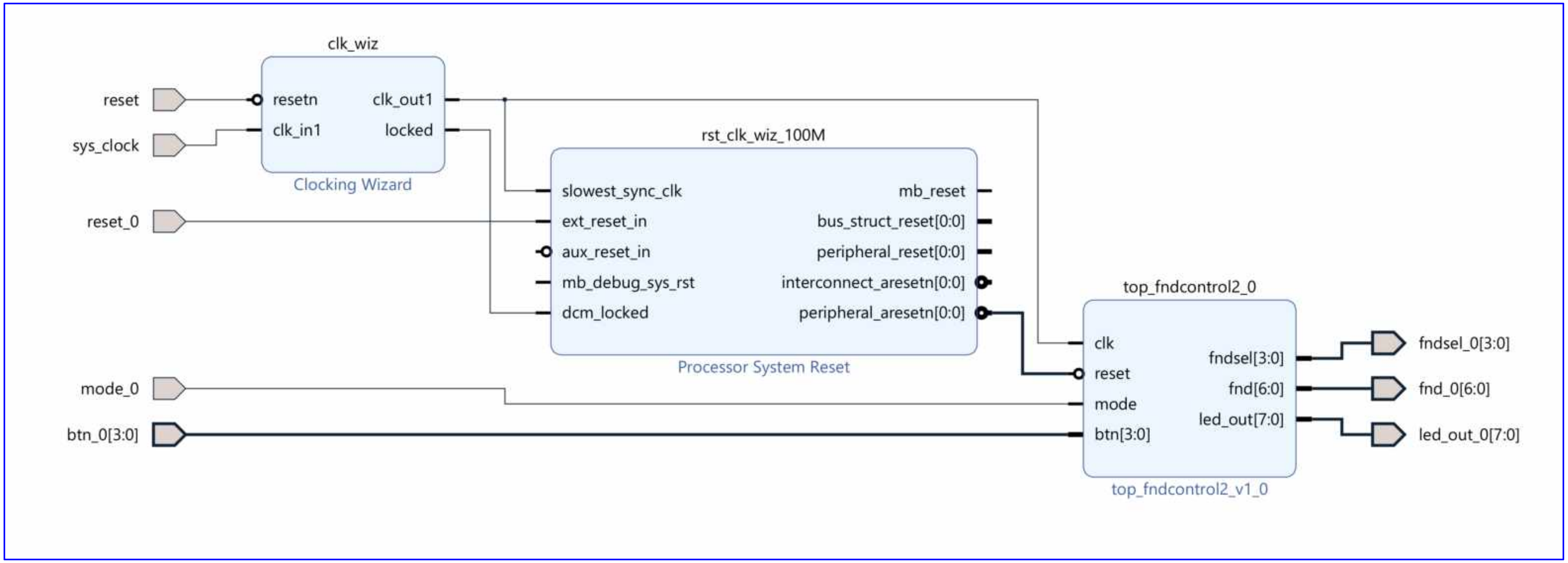
➤



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

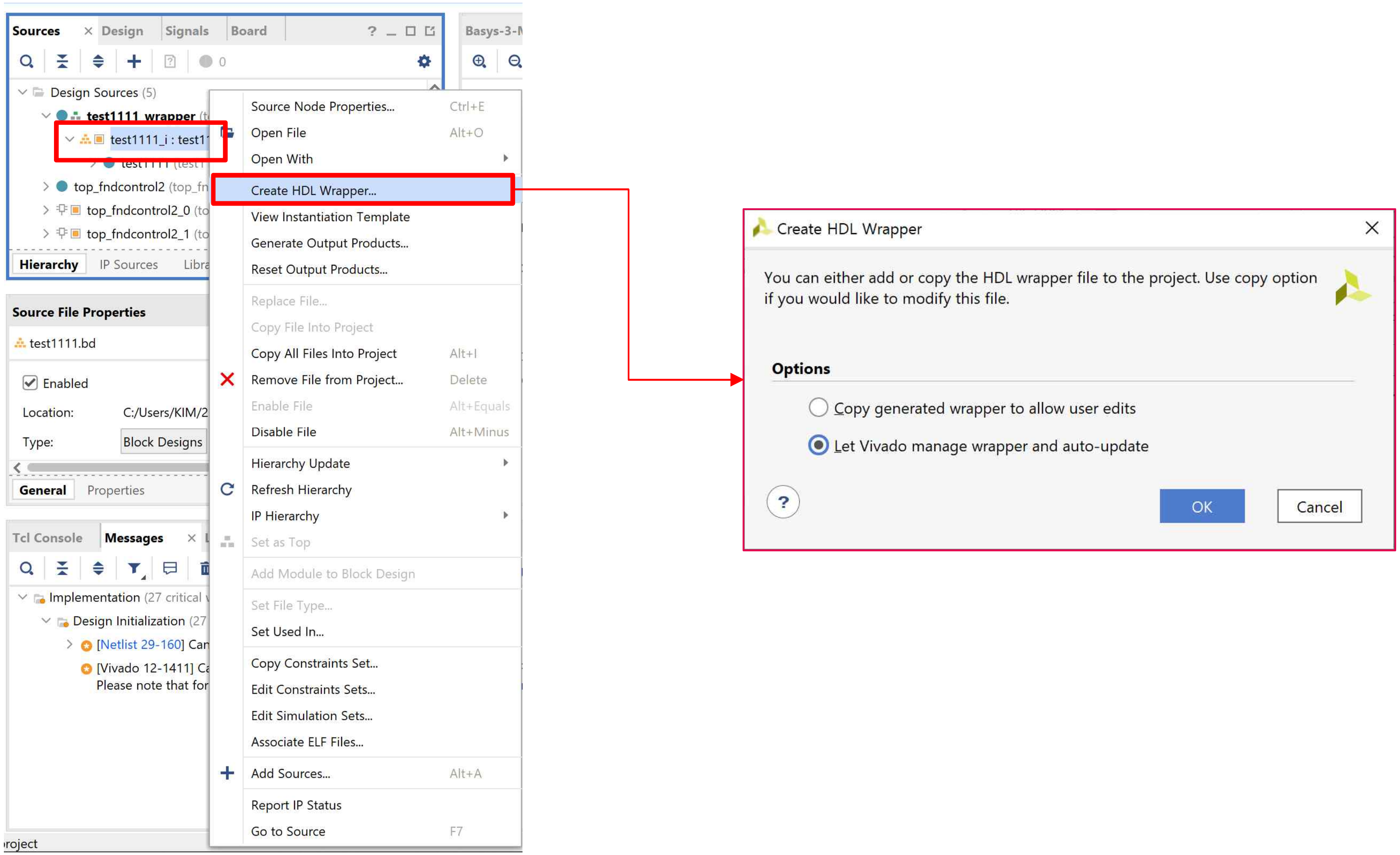
➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢ *.wrapper.v

*.wrapper.v (1)

```
`timescale 1 ps / 1 ps

module test1111_wrapper
  (btn_0,
   fnd_0,
   fndsel_0,
   led_out_0,
   mode_0,
   reset,
   reset_0,
   sys_clock);
  input [3:0]btn_0;
  output [6:0]fnd_0;
  output [3:0]fndsel_0;
  output [7:0]led_out_0;
  input mode_0;
  input reset;
  input reset_0;
  input sys_clock;
```

*.wrapper.v (2)

```
  wire [3:0]btn_0;
  wire [6:0]fnd_0;
  wire [3:0]fndsel_0;
  wire [7:0]led_out_0;
  wire mode_0;
  wire reset;
  wire reset_0;
  wire sys_clock;

  test1111 test1111_i
       (.btn_0(btn_0),
        .fnd_0(fnd_0),
        .fndsel_0(fndsel_0),
        .led_out_0(led_out_0),
        .mode_0(mode_0),
        .reset(reset),
        .reset_0(reset_0),
        .sys_clock(sys_clock));
endmodule
```

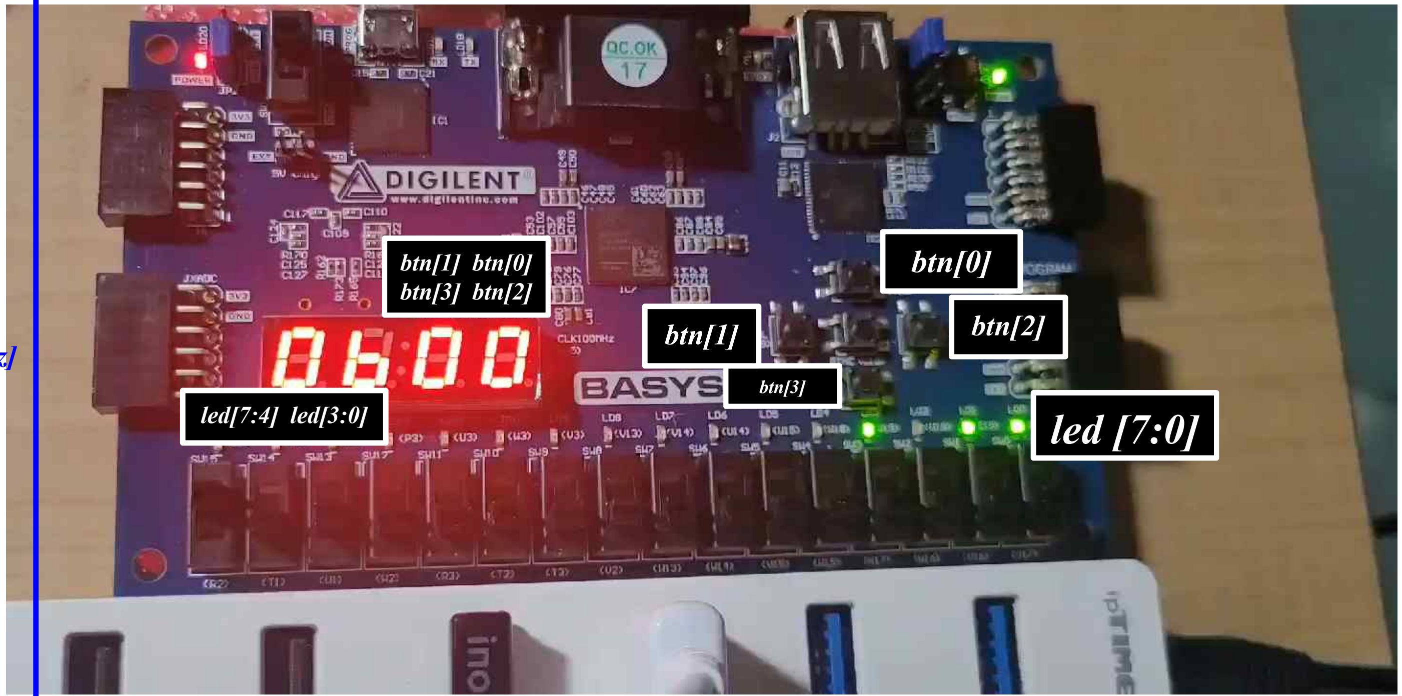
*.xdc

```
## Clock signal
set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports sys_clock]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]

## Switches
set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {reset_0}]
set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {mode_0}]
set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {reset}]

## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[0]}]
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[1]}]
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[2]}]
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[3]}]
set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[4]}]
set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[5]}]
set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[6]}]
set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led_out_0[7]}]

##7 Segment Display
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]
#set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports fnd[7]]
set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]
set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]
set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]
set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]
##Buttons
#set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports resetn]
set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports btn_0[0]]
set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports btn_0[1]]
set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports btn_0[2]]
set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports btn_0[3]]
## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
```

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

- btn_0[0] → BTNR (T18)
- btn_0 [1] → BTND (W19)
- btn_0 [2] → BTNU (T17)
- btn_0 [3] → BTNU (U17)
- reset → SW15
- mode_0 → SW14
- reset_0 → SW13
- sys_clock → System Clock [100MHz]
- FND4 → rdata 10 (led[7:4])
- FND3 → rdata 0(led[7:4])
- FND2 → Mode 0 : btn[1]
- FND1 → Mode 0 : btn[0]
- FND2 → Mode 1 : btn[3]
- FND1 → Mode 1 : btn[2]
- LD 7 ~ LD 0 → led_out [7:0]



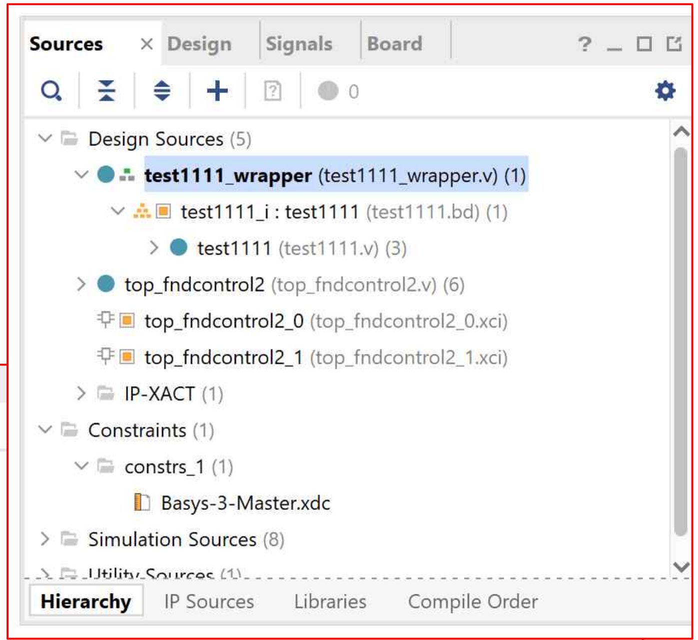
[*Create and Package New IP (II)*]

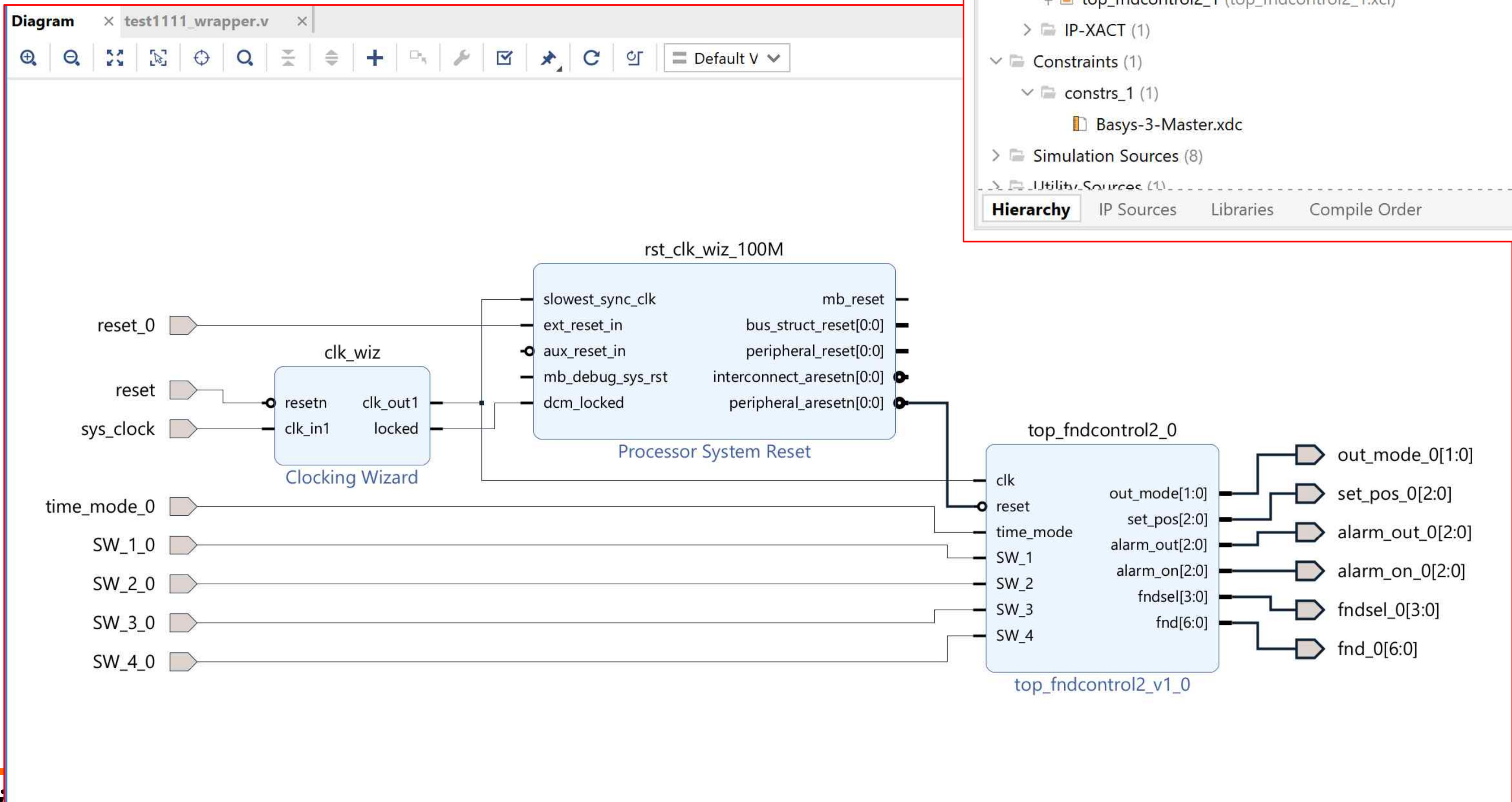
Digital_Clock and Top_FndController

➔ 실습 ⟵

Create and Package New IP :Digital Clock

- *Source and RTL*





Create and Package New IP :Digital Clock

➢ *.wrapper.v*

Diagram × Basys-3-Master.xdc × **test1111_wrapper.v** ×

C:/Users/KIM/2023/Creat_digital_clock_IP_exam/Creat_digital_clock_IP_exam.gen/sources_1/bd/test1111/hdl/test1111_wrapper.v

```verilog
`timescale 1 ps / 1 ps

module test1111_wrapper
   (SW_1_0,
    SW_2_0,
    SW_3_0,
    SW_4_0,
    alarm_on_0,
    alarm_out_0,
    fnd_0,
    fndsel_0,
    out_mode_0,
    reset,
    set_pos_0,
    sys_clock,
    time_mode_0);
  input SW_1_0;
  input SW_2_0;
  input SW_3_0;
  input SW_4_0;
  output [2:0]alarm_on_0;
  output [2:0]alarm_out_0;
  output [6:0]fnd_0;
  output [3:0]fndsel_0;
  output [1:0]out_mode_0;
  input reset;
  output [2:0]set_pos_0;
  input sys_clock;
  input time_mode_0;

  wire SW_1_0;
```

Diagram × Basys-3-Master.xdc × **test1111_wrapper.v** ×

C:/Users/KIM/2023/Creat_digital_clock_IP_exam/Creat_digital_clock_IP_exam.gen/sources_1/bd/test1111/hdl/test1111_wrapper.v

```verilog

  wire SW_1_0;
  wire SW_2_0;
  wire SW_3_0;
  wire SW_4_0;
  wire [2:0]alarm_on_0;
  wire [2:0]alarm_out_0;
  wire [6:0]fnd_0;
  wire [3:0]fndsel_0;
  wire [1:0]out_mode_0;
  wire reset;
  wire [2:0]set_pos_0;
  wire sys_clock;
  wire time_mode_0;

  test1111 test1111_i
       (.SW_1_0(SW_1_0),
        .SW_2_0(SW_2_0),
        .SW_3_0(SW_3_0),
        .SW_4_0(SW_4_0),
        .alarm_on_0(alarm_on_0),
        .alarm_out_0(alarm_out_0),
        .fnd_0(fnd_0),
        .fndsel_0(fndsel_0),
        .out_mode_0(out_mode_0),
        .reset(reset),
        .set_pos_0(set_pos_0),
        .sys_clock(sys_clock),
        .time_mode_0(time_mode_0));
endmodule

```

Create and Package New IP :Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢ .XDC

```
*.xdc

## Clock signal
set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports sys_clock]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clockk]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_1_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_2_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_3_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_4_IBUF]

## Switches
#set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {SW_1}]
#set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {SW_2}]
#set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {SW_3}]
#set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {inx[3]}]
#set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {iny[0]}]
#set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {iny[1]}]
#set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {iny[2]}]
#set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {iny[3]}]
#set_property -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {inc[0]}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {inc[1]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {inc[2]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {inc[3]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {fndin[0]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {fndin[1]}]
set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {time_mode_0}]
set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {reset}]

## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[0]}]
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[1]}]
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[2]}]
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[0]}]
set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[1]}]
set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[2]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {alarm_out[2]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {fndnumber[2]}]
set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {out_mode_0[0]}]
set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {out_mode_0[1]}]
#set_property -dict { PACKAGE_PIN W3    IOSTANDARD LVCMOS33 } [get_ports {fndnumber[5]}]
#set_property -dict { PACKAGE_PIN U3    IOSTANDARD LVCMOS33 } [get_ports {fndnumber[6]}]
#set_property -dict { PACKAGE_PIN P3    IOSTANDARD LVCMOS33 } [get_ports {fndnumber[7]}]
set_property -dict { PACKAGE_PIN N3    IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[0]}]
set_property -dict { PACKAGE_PIN P1    IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[1]}]
set_property -dict { PACKAGE_PIN L1    IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[2]}]
```

```
*.xdc

##7 Segment Display
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]

#set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports fnd[7]]

set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]
set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]
set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]
set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]

##Buttons
#set_property -dict { PACKAGE_PIN U18   IOSTANDARD LVCMOS33 } [get_ports btnC]
set_property -dict { PACKAGE_PIN T18   IOSTANDARD LVCMOS33 } [get_ports SW_3_0]
set_property -dict { PACKAGE_PIN W19   IOSTANDARD LVCMOS33 } [get_ports SW_4_0]
set_property -dict { PACKAGE_PIN T17   IOSTANDARD LVCMOS33 } [get_ports SW_1_0]
set_property -dict { PACKAGE_PIN U17   IOSTANDARD LVCMOS33 } [get_ports SW_2_0]

## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
```

Create and Package New IP :Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➢ Program Device

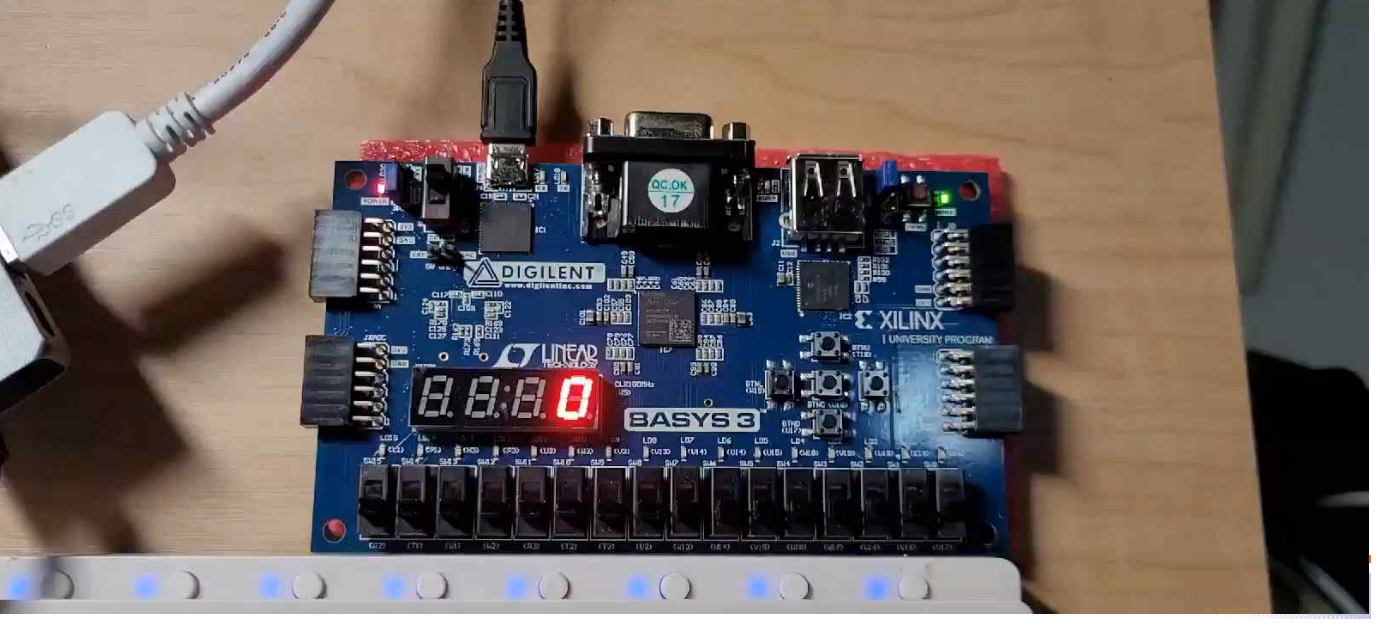
- *SW0 ➔ BTNR (T17) : Mode Selection*
- *SW1 ➔ BTND (U17) : Set_positon*
- *SW2 ➔ BTNU (T18) : Time Up*
- *SW3 ➔ BTNL (W19) : Alam On/Off*
- *reset ➔ SW15*
- *FND4 ➔ Hour 10 (or Min10)*
- *FND3 ➔ Hour 0 (or Min0)*
- *FND2 ➔ Min 10 (or Sec10)*
- *FND1 ➔ Min 0 (or Sec0)*
- *time_mode ➔ SW14 : 시분 or 분초 모드*
- *Alarm On/Off ➔ LD 0 ~ LD 2*
- *Alarm Signal ➔ LD 4 ~ LD6*
- *Mode(Selection) ➔ LD 8 ~ LD 9*
- *Set_potion ➔ LD 13 ~ LD 15*

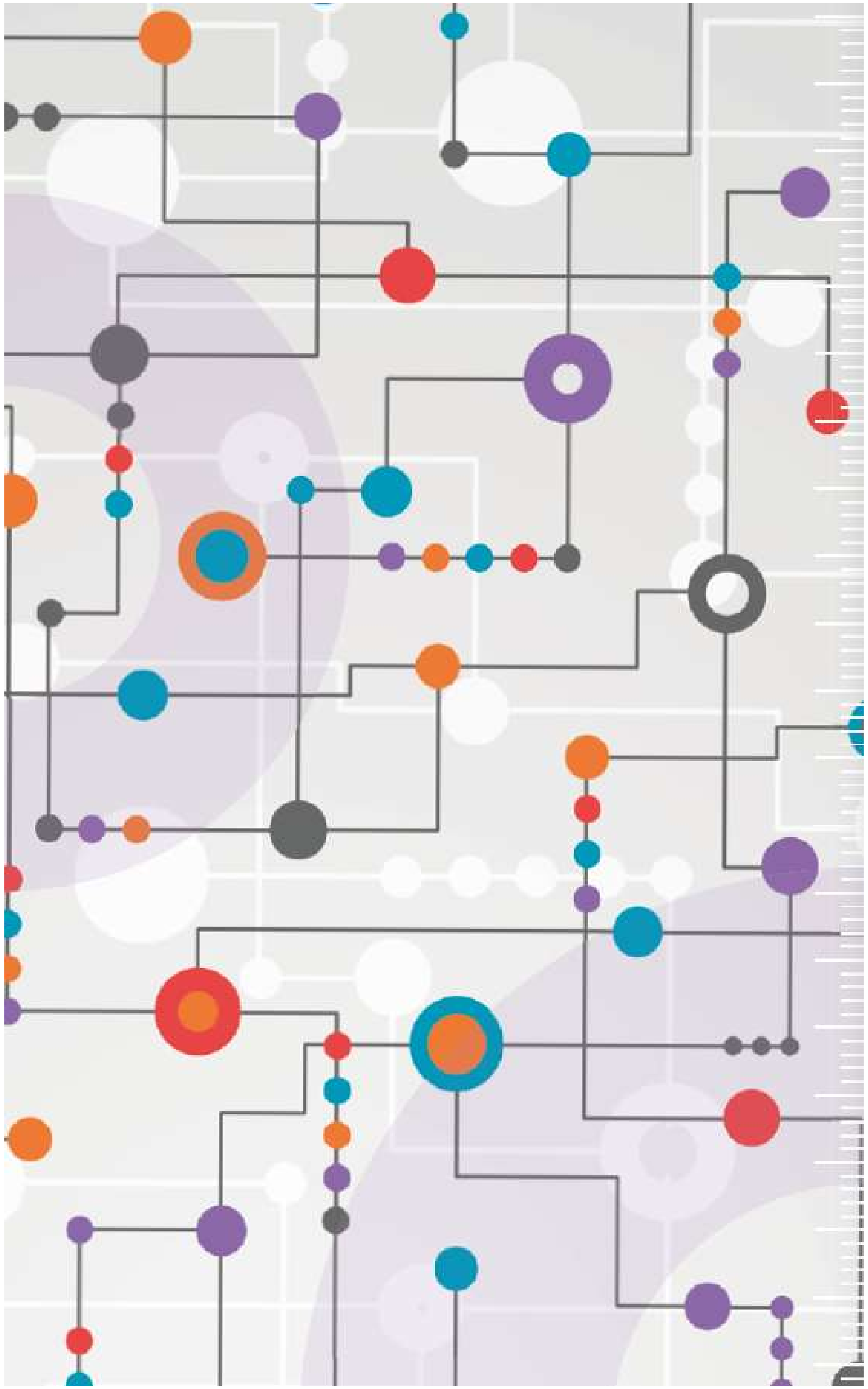
[표 6-3] 디지털 시계의 입출력 장치

입출력	입출력 장치
clk	On board 1MHz OSC
reset	리셋 스위치
SW0~SW3	입력 키 스위치 Key0~Key3
시, 분, 초	FND3~FND1
알람 ON/OFF 상태	LED D1~D3
알람 신호	LED D5~D7
모드	LED D8~D9
시, 분, 초 설정 위치 표시	LED D13~D15

[표 6-1] 디지털 시계의 스위치와 동작 모드

모드 선택(SW0)		선택 1(SW1)	선택 2(SW2)	선택 3(SW3)
0	시계 모드	현재 시간 출력		알람 ON/OFF
1	시간 설정	위치 선택 (Hour, Min)	시간 증가	
2	알람 설정	위치 선택 (Hour, Min)	시간 증가	
3	타이머(스톱워치)	스타트/스톱 (Start/Stop)	리셋	





수고하셨습니다.